KU LEUVEN

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The State of Comon's Conjecture

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Foreword

This thesis aims to document the mathematical progress made around Comon's Conjecture — a question about tensor rank and symmetry. Covered within this text are some of the attempts made to prove the conjecture in various contexts, published after it was first posited in 2008. The main focus of this work centers around the tools and methodology proposed by Shitov (both from published works [21] and preprints [22–24]) that aim to construct a valid counterexample to the conjecture. The culmination of this thesis is a close examination of Wang and Seigal's paper [26] detailing their construction of a valid counterexample in $\mathbb{R}$. While the existence of a counterexample in $\mathbb{C}$ is still an open question, this thesis concludes that the ideas detailed within the papers cited provide a valid pathway to one.

Acknowledgements

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Finally, this thesis would not exist without friends, family, coworkers, and mentors, — in Belgium and from around the world. I would not have been in any position to write this work without them.

Contents

List of Notation

General Objects

$[n]$	The set $\{1, 2, \dots, n\}$
$\mathbb{F}$	A general field, in this thesis either $\mathbb{R}$ or $\mathbb{C}$ unless specified
$\mathbb{F}[x_1, \dots, x_n]_d$	Subring of polynomial ring in n variables over $\mathbb{F}$, consisting of polynomials of fixed degree d
σ	A permutation of a finite collection of elements (the length of the collection is specified in context)

Vector Spaces

$\hat{V}_i$	The tensor product $V_1 \otimes \cdots \otimes V_{i-1} \otimes \left(\bigoplus_{\lambda \in \Lambda} \mathbb{F}\right) \otimes V_{i+1} \otimes \cdots \otimes V_d$
$\langle v_i \rangle_{i=1}^n$	The linear span of n vectors
$S^d(\mathbb{F}^n)$	Symmetric tensors (forms) of order d over $\mathbb{F}^n$
V_i	A vector space over $\mathbb{F}$ having dimension n_i
V_i^*	The dual vector space to V_i
$V_{\hat{i}}$	The tensor product of vector spaces excluding V_i, defined as $V_1 \otimes \cdots \otimes V_{i-1} \otimes V_{i+1} \otimes \cdots \otimes V_d$

Vectors

$\mathbf{1}_n$	The vector of all 1's lying in $\mathbb{F}^n$
α	A vector of non-negative integers $(a_1, \dots, a_n)$
$\mathbf{x}^\alpha$	$\prod_{i=1}^n x_i^{a_i}$ where n is taken from context
$\mathbf{x}$	A vector of indeterminates $(x_1, \dots, x_n)$
$e_{i,j}$	The j-th standard ordered basis vector in V_i. The i may not be dropped if it is clear from context
$v(k)$	The kth entry in a vector v
v_i^k	The k-th vector lying in V_i

Tensors

$\bar{\mathcal{M}}$	The tensor constructed by taking elements of $\mathcal{M}$ as slices
$\mathcal{M}$	A finite family of tensors
$T(k_1,\ldots,k_d)$	The element located at the position $(k_1,\ldots,k_d)$ in a tensor T
T^i_j	The j-th slice along the i-th mode of a tensor T (see Definition 2.10)
$T^{(i)}$	The flattening of a tensor T along the ith mode (see Definition 2.14)
$T^{\emptyset}$	The unravelling of a tensor T into a vector, with elements indexed by $\times_{i=1}^{d}[n_i]$ in lexicographic order (see remarks after Definition 2.14)
$T^{\otimes d}$	The tensor product of T with itself d times, $\underbrace{T\otimes T\otimes\cdots\otimes T}_{d\text{ times}}$

General Functions

$\mathcal{S}(T)$	The symmetrisation operator (see Definition 2.9) acting on a tensor T
$Adj(T,\mathcal{M}_1,\ldots,\mathcal{M}_d)$	The mapping denoting the general adjoining of collections of slices $\mathcal{M}_i$ along all i modes of a tensor T
$Adj^i(T,\bar{\mathcal{M}})$	The mapping denoting the adjoining of $\mathcal{M}$ as slices along the i-th mode of a tensor T (see Definition 4.2)
$C_n(T)$	The mapping that constructs the n-th clone of a tensor T
$F^{(i)}(T)$	The mapping denoting the flattening of a tensor T along the i-th mode
$Mod^i(T,\bar{\mathcal{M}})$	The mapping generating a family of tensors by slicing T with elements from $\mathcal{M}$ along the ith mode (see Definition 4.5)
$pad^i(A,T)$	The mapping denoting the padding of a lower order tensor A into the same format as T by inserting 0's along the ith mode (see Definition 4.3)
$Pad^i_j(A,T)$	The mapping taking a lower order tensor A into the same format as a higher order tensor T, now seen as the only non-zero j-th slice along the ith mode (see Definition 4.4)
$S^i_j(T)$	The mapping that selects the j-th slice along the i-th mode of a tensor T
$SAdj(T,\mathcal{M})$	The mapping denoting the symmetric adjoining operation defined for a symmetric tensor T and a collection of symmetric slices $\mathcal{M}$, $Adj(T,\mathcal{M},\ldots,\mathcal{M})$

$T \,\mathrm{mod}\, (\mathcal{M}_1, \ldots, \mathcal{M}_d)$	The mapping denoting the family generated by slicing T along each i-th mode with slices from $\mathcal{M}_i$

Scalar-Valued Functions

Rk(T)	The rank of a tensor T
SRk(T)	The symmetric rank of a symmetric tensor T

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Chapter 1

Introduction

The title of this thesis comes from an open question about tensors. Tensors are mathematical tools that are used to describe multiple linear relationships between mathematical objects. This ability of tensors to describe such inter-related information has spurred the study of *multi*-linear algebra. The focus of this thesis is threefold: how are tensors constructed, where — as mathematical objects — do they live, and how they can be broken down?

The current state of Comon's conjecture, an open problem for the past 20 years, is hard to describe succinctly. While mathematicians have steadily increased the number of cases for which the conjecture remains valid (highlighted at the end of Chapter 3), an answer to the question in full generality for $\mathbb{R}$- or $\mathbb{C}$-tensors remains unknown. The hope of this thesis is to provide a solid understanding of the work done in recent years, some of which may prove useful in constructing a counterexample to the conjecture, and to fill in the gaps present in the literature by giving formal definitions, lemmas, theorems and proofs where needed.

This question about symmetric tensors merits study because of the powerful nature of tensors as describing of multilinear relationships. One of the more important problems that has come to light in recent years is the computational complexity of matrix multiplication ([17]), and tensors and their ranks help to quantify exactly how efficient the algorithms for matrix multiplication are. Tensors also find their use in quantum mechanics (describing the entanglement of particles), general relativity (as tools

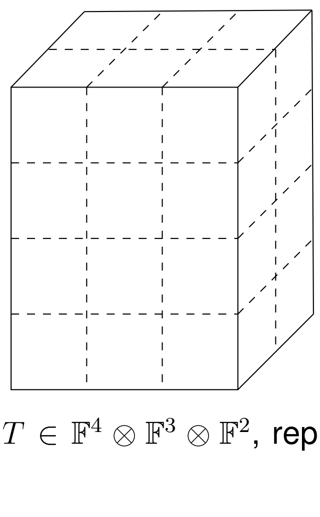

Figure 1: Example of a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, represented in the form of a 3-way array

to describe the curvature of space-time [8]), chemometrics and psychometrics (representing combinations of chemicals or traits that are interrelated [27]), in data science and computer vision [19] and in signal processing (the source selection problem [17]) — to name some more examples.

Comon's Conjecture

At its heart, Comon's Conjecture centers around two central themes: the *rank* and the *symmetry* of tensors. A tensor's rank describes how it can be decomposed into much simpler constituent tensors, and its symmetry describes how the tensor behaves under permutation actions.

The conjecture was first stated openly in July 2008 at "Geometry and Representation Theory of Tensors for Computer Science, Statistics and Other Areas" - a workshop organised at the American Institute for Mathematics [18] in the group focusing on "Multilinear Techniques in Data Analysis and Signal Processing". The conjecture as stated in the report is as follows:

Conjecture (Comon [18]). *Do the rank and symmetric rank of T always coincide, where T is a symmetric, real or complex tensor?*

This report also covered work done previously by Comon et al. [12] where they also showed that the conjecture held for conditions bounding the symmetric rank or order of a tensor in question. Since 2008, the mathematical community has further proved the conjecture true under various other assumptions.

In 2018, Shitov [21] proposed the construction of a counterexample living in $(\mathbb{C}^{800})^{\otimes 3}$ with symmetric rank at least 904, and a rank of at most 903. In 2024 however, an erratum published (Draisma [13]) pointed out an error in the original paper, rendering the conjecture still unresolved in the complex case. In the intervening time, a paper published by Wang and Seigal [26] demonstrated the construction of a real, order-6, symmetric tensor having real rank 30913 and symmetric rank 30914, thus disproving the conjecture for the real case. Their paper built on a method of construction developed in a preprint from Shitov in 2020 [22]. These ideas have also been developed further in subsequent preprints - [23] and [24], again from Shitov.

The following chapters and and subsequent sections highlight important terminology required to state and describe the setting of Comon's conjecture, review the work done to prove or disprove the conjecture in the past, and describe the major constructions required in the construction of the real counterexample.

Chapter 2: Preliminaries Regarding Tensors covers some preliminary terminology and definitions required to understand the history of attempts to prove Comon's conjecture. Personal contributions in this chapter include the proof of a theorem showing the isomorphism between symmetric tensors of fixed order and homogenous polynomials of fixed degree, Lemma 2.2, Definition 2.11, Definition 2.12, Definition 2.13 and Definition 2.15.

Chapter 3: Literature Review reviews the literature that currently exists since the statement of the problem in 2008. The papers discussed highlight the valid attempts at proving Comon's conjecture over $\mathbb{R}$ and $\mathbb{C}$, and the attempts to construct a counterexample. Personal contributions to this chapter are the compilation of all these results in a singular location, highlighting the state of the art currently.

Chapter 4: Constructions Required for a Counterexample focuses on understanding Shitov's methodology and approach to constructing a counterexample to Comon's conjecture. The culmination of this chapter is an examination of Wang and Seigal's publication [26], which presents real-valued counterexample to the conjecture. Personal contributions to this chapter include explicit proofs for the various operations discussed, most of which do not exist in the existing literature.

Chapter 5: Conclusion illustrates problems with the current approach to constructing counterexamples and seeks to highlight potential future avenues to solving this problem once and for all.

Further personal contributions to this thesis include all the illustrations and examples, used to shed light on definitions given through the entire document.

Chapter 2

Preliminaries Regarding Tensors

2.1 Tensors in an Algebraic Setting

The most abstract way of thinking about tensors is when describing them as elements of a tensor product of modules. Consider A, B and P as being modules over a ring R. Then any bi-linear mapping, $t : A \times B \to P$, is described by acting on elements from $A \times B$ in the following manner:

$$t(\lambda a, b) = t(a, \lambda b) = \lambda t(a, b), a \in A, b \in B, \lambda \in R, t(a, b) \in P$$

$$\begin{tikzcd}
A \times B \arrow[rr, "t \text{ bilinear}=\tilde{t}\circ\varphi"] \arrow[dr, "\varphi"'] & & P \\
& A \otimes_R B \arrow[ur, "\exists! \tilde{t}"'] &
\end{tikzcd}$$

Figure 2: Illustrating a tensor product of R-modules

Such a bilinear, or even generalising to a multi-linear, mapping between R-modules can be represented by an element lying in their tensor product, such that Figure 2 commutes [25] (illustrated specifically for the case mentioned above). Such a tensor product of R-modules is unique upto unique isomorphisms, and this is referred to in algebraic literature as a *universal property* of tensor products.

While such an abstract description is useful when showing that any general elements lying in the specified spaces A and B can be combined to form a tensor, it does not specify details required for showing how this tensor can be represented.

A more useful approach to understanding tensors is to begin from a foundation in linear algebra. As matrices represent linear transformations between vector spaces V and W, so d-way arrays can be used to represent multilinear relationships between vector spaces $V_1, V_2 \ldots V_d$, where each V_i has a dimension n_i. Diving deeper into this analogy, a matrix of the form $n_i \times n_j$ behaves either as a bilinear map from the product $V_i^* \times V_j \to \mathbb{F}$, or as linear map $V_j \to V_i$ or as its dual $V_i^* \to V_j^*$. This hybrid nature of matrices is a reflection of the underlying fact that matrices represent order-2 tensors. Since computing elements A_{ij} of the matrix A depends on a choice of bases for both vector spaces V and W, an analogous choice of bases must be made when representing a d-linear tensor, $T \in V_1 \otimes V_2 \otimes \cdots \otimes V_d$, as a d-way array.

For the purposes of this thesis, unless specified, each vector space will be considered over a field $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Here each vector space V_i is equipped with its standard basis, $\{e_{i,j}\}_{j=1}^{n_i}$, where $e_{i,j}(k) = \delta_{j,k}$. Each different choice of basis changes how the tensor T is represented as a d-way array.

Definition 2.1 (Order of a Tensor)**.** An element lying in the tensor product of d vector spaces, $T \in V_1 \otimes \cdots \otimes V_d$, is said to be an **order-d tensor**.

Consider $T := v_1 \otimes \cdots \otimes v_d$, a specific order-$d$ tensor, with each component vector $v_i \in V_i$. Then, in coordinates

$$v_i = \sum_{j=1}^{n_i} v_i(j) e_{i,j},\ v_i(j) \in \mathbb{F}\ \forall j = 1, \ldots, n_i.$$

The elements of a d-way array that corresponds to T then can be given in the following manner:

$$T(k_1, k_2, \ldots, k_d) = \prod_{i=1}^{d} v_i(k_i),$$

where $0 \leq k_i \leq n_i$.

In a similar fashion, given two tensors in $V_1 \otimes \cdots \otimes V_d$ of the specific form $T_1 := v_1 \otimes \cdots \otimes v_d$ and $T_2 := w_1 \otimes \cdots \otimes w_d$, then one can describe the d-way array corresponding to the tensor $T_1 + T_2$ in coordinates as follows

$$(T_1 + T_2)(k_1, k_2, \ldots, k_d) = \left(\prod_{i=1}^{d} v_i(k_i) \right) + \left(\prod_{i=1}^{d} w_i(k_i) \right),$$

indicating simply a componentwise addition in the corresponding d-way arrays. A similar description can be made when scaling a tensor by any factor $\lambda \in \mathbb{F}$ — λT.

Therefore an order d tensor $T \in V_1 \otimes \cdots \otimes V_d$ that can be expressed as a sum of these specific *elementary* forms, as above, can also be represented by a d-way array. This property is satisfied by any general order-d tensor (see Definition 2.5).

Definition 2.2 (Modes of a Tensor)**.** Given an order-d tensor, $T \in V_1 \otimes \cdots \otimes V_d$, the modes of T refer to specifying the d distinct ways that T as a tensor can be oriented. Thus a tensor of order n has exactly n modes.

Remark. Modes of are useful because they provide a specific orientation along which operations can be performed on a given tensor. When working with tensors as d-way arrays, the modes can be seen as the distinct indices in the overall tuples that index the elements of the array.

Example 2.1. Consider

$$T := (a, b, c, d) \otimes (x, y, z) \otimes (p, q) \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2.$$

Then T is an order-3 tensor having 3 modes.each entry in the 3-way array representing T is indexed by a tuple $(i_1, i_2, i_3) \in [4] \times [3] \times [2]$.

Definition 2.3 (Format of a Tensor)**.** For a given tensor $T \in V_1 \otimes \cdots \otimes V_d$, the format of T refers to the tuple $(n_1, \ldots, n_d)$, conveying all the possible indices of the d-way array that represents T. For any two tensors lying in the same tensor space, their formats are the same. Here, the n_i are exactly the dimensions of V_i.

Viewing Tensors

A tensor $T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$ can be viewed in the following different ways [17]:

1. As a multilinear mapping between $V_1^* \times V_2^* \times \cdots \times V_d^* \to \mathbb{F}$ (extending the analogy of looking at matrices as order-2 tensors)

$$(f_1, f_2 \ldots f_d) \mapsto \prod_{i=1}^{d} f_i(v_i).$$

   Here $f_i \in V_i^*$, the dual space to each vector space V_i.

2. For every choice of $i \in \{1, \ldots, d\}$, T can be viewed as a linear mapping from $V_i^* \to V_{\hat{i}} := V_1 \otimes \cdots \otimes V_{i-1} \otimes V_{i+1} \otimes \cdots \otimes V_d$. Viewing T this way takes a vector (an order-1 tensor) to an order-$(d-1)$ tensor.

$$f_j \mapsto f_j(v_i)(v_1 \otimes \cdots \otimes v_{i-1} \otimes v_{i+1} \otimes \cdots \otimes v_d). \tag{2.1}$$

   This viewpoint can be seen when defining an operation that selects a specific "slice" of a tensor (or even a linear combination of them), after fixing the mode i.

3. Finally, as the dual of the defined linear map (see Equation (2.1)), now going from $V_{\hat{i}}^* \to V_i$. This point of view of transfroms an order-$(d-1)$ tensor to a rescaling of the vector v_i.

$$f_1 \otimes \cdots \otimes f_{i-1} \otimes f_{i+1} \otimes \cdots \otimes f_d \mapsto \left(\prod_{j \neq i} f_j(v_j) \right) v_i.$$

   Here note that the dual operation and the tensor product operation commute.

Each of these different viewpoints holds for order-d tensors in general, not just for these specific elementary tensors. This becomes evident in the following section on tensor rank and decompositions.

2.2 Rank

The first key focus of Comon's conjecture involves figuring out how tensors can be built. The following example shows that there exist tensors that cannot be written as simply the tensor product of vectors analogously to how not all vectors in a vector space can be written as rescaled basis vectors.

Example 2.2. When dealing with 2×2 matrices $\in \mathbb{F}^2 \otimes \mathbb{F}^2$, consider $e_1 \otimes e_1 + e_2 \otimes e_2$ (the identity matrix). Such a matrix cannot be written as $v_1 \otimes v_2$, a simple tensor product of two vectors, where $v_1, v_2 \in \mathbb{F}^2$.

Making this preliminary observation leads to the following definition —

Definition 2.4 (Elementary Tensor)**.** Consider an order-d tensor, T, of the specific form $T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$. Since T can be written explicitly as the tensor product of d vectors, it is called an **elementary tensor**. Equivalent names for such tensors in the literature reviewed are *rank*-1 tensors and *decomposable* tensors.

Using these elementary tensors as building blocks provides a way to describe the construction of any general order-d tensors. In the most naive way, this can be done by considering all elementary tensors of the form $e_{1,j_1} \otimes \cdots \otimes e_{d,j_d}$. This explains why any general order-d tensor can be represented by a d-way array, and also why the three viewpoints specified above for an elementary tensor can be extended to a general order-d tensor.

Definition 2.5 (Tensor Decomposition [27])**.** Consider any order-d tensor $T \in V_1 \otimes \cdots \otimes V_d$. A **decomposition** of T consists of writing T as a sum of elementary tensors

$$T = \sum_{k=1}^{s} v_1^k \otimes \cdots \otimes v_d^k,$$

with each of the $v_i^k \in V_i$.

Remark. Here note that there exist multiple names for describing the decomposition of a tensor into the sum of elementary tensors - polyadic forms, canonical decomposition (CANDECOMP), paralell factors (PARAFAC) etc. The above definition agrees with what is known in the literature as CP-decomposition (a combination of names CANDECOMP/PARAFAC). [15]

Thus, for any given tensor T, there can exist multiple decompositions, not just depending on the choice of basis vectors for each V_i. Two questions naturally follow this definition of tensor decomposition:

Q1. For a given order-d tensor T, does there exist a shortest decomposition?

Q2. If such a shortest decomposition exists, is it unique (upto the choice of a different set of basis vectors)?

The answer to the first question can be given in terms of the following definition:

Definition 2.6 (Tensor Rank)**.** The **rank** of a tensor T, denoted **Rk**(T), is the minimal length of any possible decomposition of T.

A rank decomposition of a tensor always exists, because tensors can always be written in terms of elementary tensors. Computing the rank of a given tensor is not an easy problem, but there exist multiple approaches in the literature that rely on both the algebraic properties of tensors and associated transformations, and the geometric properties of the varieties on which these tensors lie. Tensor rank also does not depend on which elementary tensors are involved in a decomposition. Thus Q1 as given above has an affirmative answer, but Q2 does not — multiple rank decompositions can exist for a specified tensor..

2.3 Symmetric Tensors

The other focus of Comon's Conjecture revolves around tensors that exhibit the property of symmetry. These tensors become valuable in applications simply because the symmetry means that they require "less" information to be constructed. What makes a tensor symmetric is the next logical question to tackle, and it becomes necessary to define how permutations act on a given tensor.

Given $\sigma \in S_d$ (the symmetry group of $\{1, 2, \ldots, d\}$), the permutation action of σ on a tensor T can be defined as follows:

Definition 2.7 (Permutation of a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be an order-d tensor. Then $\sigma \in S_d$ acts on all tensors in V by linearly extending the following mapping of elementary tensors:

$$\sigma : V_1 \otimes \cdots \otimes V_d \to V_{\sigma_1} \otimes \cdots \otimes V_{\sigma_d}, v_1 \otimes \cdots \otimes v_d \to v_{\sigma_1} \otimes \cdots \otimes v_{\sigma_d}.$$

This induces a corresponding mapping from a general $T \mapsto \sigma \cdot T := \sigma \cdot (\sum_{i=1}^{r} T_i)$, where each T_i is an elementary tensor.

To talk about tensors being invariant under permutation actions, it is necessary to work with tensors that retain the same format after permutation. This means symmetric tensors can only exist in the space $V^{\otimes d} := \underbrace{V \otimes \cdots \otimes V}_{d \text{ times}}$. These tensors are sometimes referred to in the literature as being *cubical*.

The definition of a symmetric tensor now follows:

Definition 2.8 (Symmetric Tensor [12])**.** Consider a tensor $T \in V^{\otimes d}$. T is said to be **symmetric** if the following is true:

$$T = \sigma \cdot T \; \forall \; \sigma \in S_d. \tag{2.2}$$

Remark. As a consequence of this definition, the following equivalent equality condition also holds for symmetric tensors

$$T = \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot T.$$

This gives rise to an operator that transforms any tensor into a symmetric tensor.

Definition 2.9 (Symmetrisation Operator)**.** Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order-d tensor. Then the **symmetrisation operator** is defined as follows:

$$\mathcal{S} : V \to V, \; T \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot T. \tag{2.3}$$

The resulting tensor $\mathcal{S}(T)$ is always symmetric.

Since every tensor can be expressed as a sum of elementary tensors (see Definition 2.5), **Rk**(T) is defined for symmetric tensors T as well. However, one can define *elementary symmetric tensors* as tensors in $V^{\otimes d}$ of the form $v^{\otimes d}$ where $v \in V$. Establishing this definition allows for symmetric tensors to be written as a sum of these elementary symmetric tensors, resulting in the definition of **symmetric decompositions** and the analogous notion of **symmetric rank**, denoted **SRk**(T). The following lemma (Lemma 4.2 in [12]) shows why such a symmetric decomposition always exists for a symmetric tensor T.

Lemma 2.1 ([12]). *Let T be a symmetric, order-d tensor lying in $(\mathbb{F}^n)^{\otimes d}$. Then there exist vectors $v_1, \ldots, v_s \in \mathbb{F}^n$ such that*

$$T = \sum_{i=1}^{s} v_i^{\otimes d}. \tag{2.4}$$

Proof. Suppose the lemma is false. Then there must exist a symmetric, order-d tensor T that cannot be expressed in the form $\sum_{i=1}^{s} v_i^{\otimes d}$ for any $s > 0$. Then T can only be expressed as an infinite sum of symmetric tensors. Since the space $V_1 \otimes \cdots \otimes V_d$ is finite dimensional, such a sum can be expressed in a finite manner, leading to a contradiction of the initial assumption that the lemma was false. Therefore, a symmetric decomposition must always exist for T, a symmetric order-d tensor. $\square$

Remark. The proof given in [12] involves showing that the space generated by all k-th tensor powers of linear forms in the symmetric tensors cannot be contained in any non-zero non-trivial hyperplane. Their proof also involves defining a bilinear form on the space of the symmetric tensors.

Example 2.3. Consider the tensor

$$\begin{aligned} T &:= (1,0,0) \otimes (1,0,0) \otimes (1,0,0) + (0,0,1) \otimes (0,0,1) \otimes (0,0,1) \\ &\quad +(0,1,0) \otimes (0,1,0) \otimes (0,1,0) \\ &= (1,0,0)^{\otimes 3} + (0,0,1)^{\otimes 3} + (0,1,0)^{\otimes 3}. \end{aligned}$$

This is definitely a symmetric decomposition, and T is a symmetric tensor.

It is clear that not every tensor can be seen as symmetric, and therefore the subset of all order-d tensors that are symmetric is denoted $S^d(\mathbb{F}^n) \subsetneq (\mathbb{F}^n)^{\otimes d}$ — notation usually reserved for homogenous polynomials in n variables of fixed degree d. In fact it can be shown that these spaces, $S^d(\mathbb{F}^n)$ and $\mathbb{F}[x_1, \ldots, x_n]_d$, are isomorphic to each other. This isomorphism is a powerful tool that allows for an interchange of polynomial and tensor notation (in an appropriate manner) when dealing with symmetric tensors.

Symmetric Tensors and Homogenous Polynomials of Fixed Degree

A homogenous polynomial $f \in \mathbb{F}[x_1, \ldots, x_n]_d$ is of the form

$$f(x_1, \ldots x_n) = \sum_{a_i \geq 0, \sum_i^n a_i = d} \lambda_{(a_1, \ldots, a_n)} \prod_{i=1}^{n} x_i^{a_i}, \tag{2.5}$$

where all the coefficients $\lambda_{(a_1, \ldots, a_n)} \in \mathbb{F}$. The following notation is also standard in the literature surveyed, taking $\mathbf{x} := (x_1, x_2, \ldots x_n)$ to be the tuple of variables and the tuple of exponents denoted $\alpha := (a_1, a_2, \ldots a_n) \in (\mathbb{N} \cup \{0\})^n$. Then one has $\mathbf{x}^\alpha := \prod_{i=1}^{n} x_i^{a_i}$.

Further $|\alpha| = \sum_i^n a_i$, so the homogenous polynomial in Equation (2.5) can be rewritten as

$$f(\mathbf{x}) = \sum_{|\alpha|=d} \lambda_\alpha \mathbf{x}^\alpha. \tag{2.6}$$

The following proposition is standard in tensor decomposition literature, but is stated again for convenience.

Proposition 2.1. *For fixed values of $d, n > 0$, the map*

$$\Phi_d : \mathbb{F}[x_1, \dots, x_n]_d \to S^d(\mathbb{F}^n),$$

defined by linearly extending the following mapping of the degree-d monomials

$$\mathbf{x}^\alpha \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right) = \mathcal{S} \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right), \tag{2.7}$$

is an isomorphism of vector spaces over $\mathbb{F}$.

Proof. Each degree-d monomial, $\mathbf{x}^\alpha$, can be seen as a basis vector in the vector space $\mathbb{F}[x_1, \dots, x_n]_d \cong \mathbb{F}^{\binom{n+d-1}{d}}$. These monomials are mapped to symmetric tensors, by the definition of the symmetrisation operator (see Definition 2.9). Φ_d is thus a well-defined mapping, since all other homogenous degree-d polynomials can be constructed as linear combinations of these basis vectors. Therefore for $\lambda \in \mathbb{F}$,

$$\Phi(\lambda \mathbf{x}^\alpha + \mathbf{x}^\beta) = \lambda \left(\frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{\otimes a_i}) \right) + \left(\frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{\otimes b_i}) \right) = \lambda \Phi(\mathbf{x}^\alpha) + \Phi(\mathbf{x}^\beta).$$

Further, every elementary symmetric tensor $v^{\otimes d} = (\sum_{i=1}^n v_i e_i)^{\otimes d}$ can be seen as the image of the d-th power of the linear form $(\sum_{i=1}^n v_i x_i)^d$. Since every symmetric tensor has a symmetric tensor decomposition, Φ_d must also be surjective.

Suppose Φ_d is not injective. Then there exist $f, g \in \mathbb{F}[x_1, \dots, x_n]_d$ such that $\Phi_d(f) = \Phi_d(g)$ but $f \neq g$. Now if $f \neq g$, then there is at least one coefficient α' such that $f_{\alpha'} \neq g_{\alpha'}$. Without a loss of generality, let this be the only index where the functions are not equal. Now we definitely have that $\Phi_d(f) - \Phi_d(g) = 0$, by our assumption. This is equivalent to the following

$$\sum_{|\alpha|=d} \frac{f_\alpha - g_\alpha}{d!} \sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right) = 0.$$

But this simplifies to the following expression

$$\frac{f_{\alpha'} - g_{\alpha'}}{d!} \left(\sum_{\sigma \in S_d} \sigma \cdot \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right) \right) = 0,$$

a contradiction, because none of the tensors on the left side of the multiplication are 0, and by the initial assumption, the fraction must also be non-zero, and this multiplication is occurring in an integral domain. Thus the initial assumption must be false, and Φ_d must be injective. □

In fact Φ_d can be seen as a component of a larger isomorphism between graded vector spaces, $\Phi : \oplus_{i \geq 0} \mathbb{F}[x_1, \dots, x_n]_i \to \bigoplus_{d \geq 0} S^d(\mathbb{F}^n)$ [12].

2.4 Working with Tensors

The questions of how tensors are constructed and where they live have been addressed in the preceding sections. The goal of this final section of the chapter is to provide the necessary tools and terminology required when performing tensor analysis.

2.4.1 Slicing a Tensor

Viewing the columns or rows of a matrix as linearly dependent or independent vectors and how they affect the overall linear transformation described by the matrix is the exact low-order analogue for describing tensors as composed of lower-order slices. Conventionally the columns of a matrix would be the 1-slices, since they occur along the "first" mode, and the rows would therefore be considered 2-slices.

Definition 2.10 (Slices of a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be a general order-d tensor. Then T admits a rank decomposition

$$T := \sum_{i=1}^{r} v_1^i \otimes \cdots \otimes v_d^i,\ v_j^i \in V_j\ \forall\ j = 1, \ldots, d.$$

Then the j**-th slice along the** k**-th mode** (also called the j-th k-slice) is an order-$(d-1)$ tensor, lying in $V_{\hat{k}}$, given by the following formula:

$$T_j^k := \sum_{i=1}^{r} v_k(j)(v_1^i \otimes \cdots \otimes v_{k-1}^i \otimes v_{k+1}^i \otimes \cdots \otimes v_d^i).$$

In this definition, $1 \leq j \leq n_k$ since these slices are indexed upto the dimension of V_k.

When viewing T as a multi-way array, the j-th k-slice is equivalent to working with the $(d-1)$-way array, generated by fixing the k-th index in the original d-way array.

$$T_j^k(\lambda_1, \ldots, \lambda_{k-1}, \lambda_{k+1}, \lambda_d) = T(\lambda_1, \ldots, \lambda_{k-1}, j, \lambda_{k+1}, \ldots, \lambda_d)$$

Observation. This definition of the j-th k-slice cannot be made coordinate independent. The various k-slices are differentiated by the fact that as a vector $v_k^i = (v_k^i(j))_{j=1}^{n_k}$, and this representation depends on a choice of basis for V_k. Since $V_k \cong \mathbb{F}^{n_k}$, unless another basis specified, the standard basis $\{e_j\}_{j=1}^{n_k}$ is used.

Remark. Through the rest of this thesis, the superscript accompanying a tensor variable, written in uppercase roman letters, indicates the chosen mode along which a specified operation must occur.

Example 2.4. Consider a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, where $T := (1,2,3,4) \otimes (1,2,3) \otimes (1,2)$. Then slices along different modes are computed as follows (visualised in Figure 3)

- $T_3^1 = 3((1,2,3) \otimes (1,2))$
- $T_2^2 = 2((1,2,3,4) \otimes (1,2))$

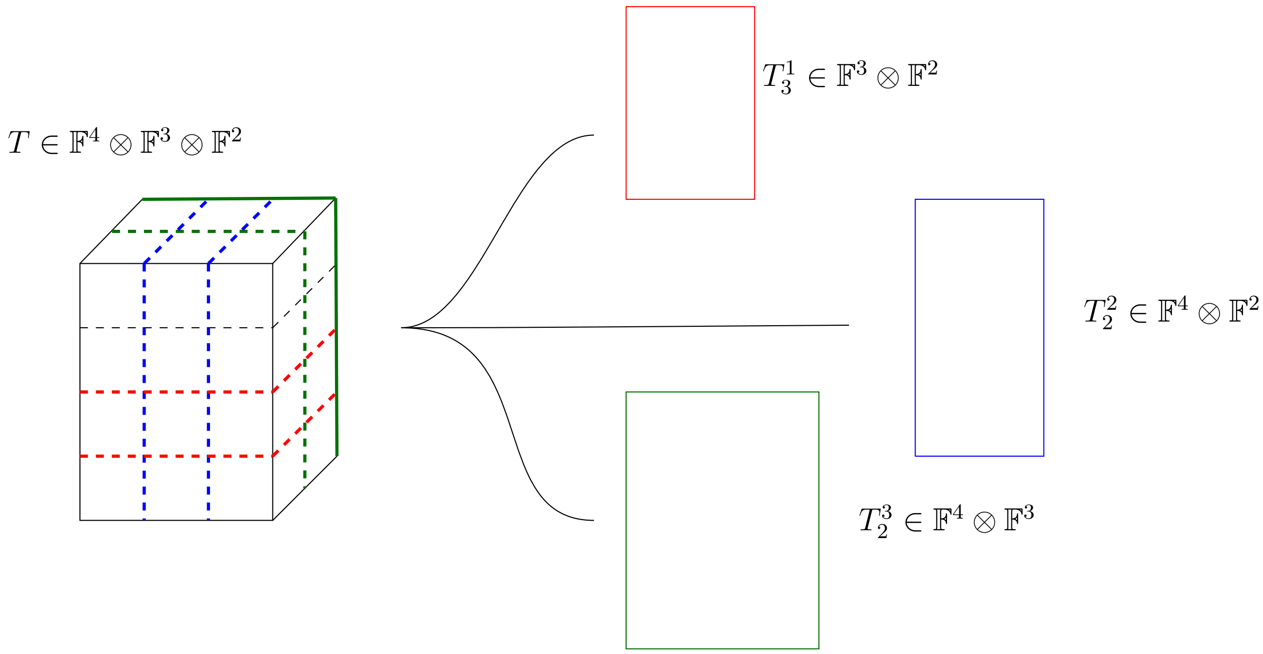

Figure 3: Slicing a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$ along different modes

- $T_2^3 = 2((1, 2, 3, 4) \otimes (1, 2, 3))$

Definition 2.11 (Slicing Operator)**.** Given the space $V_1 \otimes \cdots \otimes V_d$, and integers $1 \leq k \leq d$ and $1 \leq j \leq n_k$, define the operation

$$S_j^k : V_1 \otimes \cdots \otimes V_d \to V_{\hat{k}}, T \mapsto T_j^k.$$

Such an operator is well-defined, and is, in fact, a linear operator.

Lemma 2.2. *Given a tensor $T \in V_1 \otimes \cdots \otimes V_d$, such that **Rk**$(T) = r$, then there is at least one mode $1 \leq i \leq d$ having r linearly independent slices T_j^i where $1 \leq j \leq n_i$.*

Proof. Suppose the claim is false i.e. there is no mode i such that the number of linearly independent slices T_j^i is r. Since we are only dealing with a finite order d, there is a mode i with the largest number of linearly independent slices T_j^i. Let this number be ρ. By the initial supposition, one must have:

$\rho < r$**:** Each of the slices $S_i^j(T) = T_j^i$ can be written as a linear combination of exactly ρ slices of order $(d-1)$. One can therefore rewrite the decomposition of T in terms of vectors in V_i and the linearly independent set of $S_j^i(T)$, a decomposition of length less than r, contradicting the definition of $r =$ **Rk**(T).

$\rho > r$**:** Then any decomposition of T with length r must miss out at least one of the linearly independent slices that make up T, a contradiction to the initial idea of decomposing T.

Thus the assumption of the claim being false leads to a contradiction in either case. Therefore $\rho = r$, for at least one mode i. $\square$

A major claim from the literature on Comon's conjecture is the following theorem about the number of linearly independent slices along a chosen mode, when working with rank decompositions. The theorem as stated below first appeared as Lemma 3.1 in [27], but in a more geometric formulation it was proved earlier as Proposition 2.4 in [7]. The proof given below seeks a balance between these two formulations.

Theorem 2.1 ([27], [7]). *Let T be an order-d tensor of rank r. Then a rank decomposition of T is given as*

$$T := \sum_{k=1}^{r} v_1^k \otimes \cdots \otimes v_d^k.$$

For any mode i, the set of slices of the decomposition

$$\{v_1^k \otimes \cdots \otimes v_{i-1}^k \otimes v_{i+1}^k \otimes \cdots \otimes v_d^k \in V_{\hat{i}} \mid 1 \leq k \leq r\}$$

is linearly independent.

Proof. Let $1 \leq i \leq d$ be fixed, as an arbitrary mode for this proof. Then the tensor $T \in V_i \otimes V_{\hat{i}}$. Define a curve

$$S(t) := \sum_{k=1}^{r} v_i^k(t) \otimes v_{\hat{i}}^k(t),\ S(0) = T. \tag{2.8}$$

Here $v_i^k(t)$ are curves in V_i, and $v_{\hat{i}}^k(t)$ are curves in $V_{\hat{k}}$. One can restrict to working with $V_i' \subset V_i$ instead of V_i, while keeping $S(t) \in V_i' \otimes V_{\hat{i}}$. This can also be followed by assuming that $S(t) : V_i^* \to V_{\hat{i}}$ an injective mapping, when t is sufficiently close to 0. Then since $T(V_i^*)$ can be written as $\lim_{t\to 0} S(t)(V_i^*) \subseteq \lim_{t\to 0}\langle v_{\hat{i}}^k(t)\rangle_{k=1}^r$, one has **Rk**$(T(V_i^*)) \leq$ **Rk**(T).

Conversely, consider a set $w_{\hat{i}}^k(t) \subset V_{\hat{i}}$ such that $T(V_i^*) \subset \lim_{t\to 0}\langle w_{\hat{i}}^k\rangle_{k=1}^r$, where the dimension of the span of $w_{\hat{i}}^k(t)$ is assumed to be constant. Then, considering a basis $\{f_{i,j}\}_{j=1}^{n_i}$ of V_i^* (dual to the basis $\{e_{i,j}\}_{j=1}^{n_i} \subset V_i$), one can write

$$T(f_{i,j}) = \lim_{t\to 0} \sum_{k=1}^{r} \lambda_{j,k} w_{\hat{i}}^k(t),$$

for functions $\lambda_{j,k}(t)$. Then the tensor T can be written as

$$T = \sum_{j=1}^{n_i} e_i^j \otimes \left[\lim_{t\to 0} \sum_{k=1}^{r} \lambda_{j,k}(t) w_{\hat{i}}^k(t)\right].$$

Here the $\lambda_{j,k}(t)$ are appropriately chosen functions. By the multilinear nature of tensor products, this can be rewritten as

$$S(t) = \sum_{k=1}^{r}\left[\sum_{j=1}^{n_i} \lambda_{j,k} e_i^j(t)\right] \otimes w_{\hat{i}}^k(t).$$

At $t \to 0$, this forms a decomposition of T which must necessarily be at least as long as the rank decomposition of T. Thus **Rk**$(T) \leq$ **Rk**$(T(V_i^*))$. $\square$

Remark. When working with $T \in V_1 \otimes \cdots \otimes V_d$, a general tensor, then the format of $S_j^k(T)$ - the j-th k-slice of T is a tuple of length $d-1$, contained in the format of T.

Remark. When working with symmetric tensors $T \in S^d(V) \subset V^{\otimes d}$ the operation of taking slices $S_j^{k_1}(T) = S_j^{k_2}(T)$ for all $1 \leq k_1 \leq k_2 \leq d$ and a fixed $0 < j \leq \dim V$. This is by the symmetric nature of T. Thus when working with symmetric tensors, slicing operations may be specified along any mode.

In [26], the authors also describe the linear spaces generated by a family of slices in the following manner:

Definition 2.12 (i-th slice space of T)**.** For a given tensor $T \in V_1 \otimes \cdots \otimes V_d$, the *$i$-th slice space of* T, denoted $\mathcal{L}_i(T)$, is defined:

$$\mathcal{L}_i := \langle T_j^i \rangle_{j=1}^{n_1} \subseteq V_{\hat{i}}, \tag{2.9}$$

or the linear span generated by all the i-slices of T.

2.4.2 Flattening a Tensor

Flattening, while not as important in Shitov's constructions [22–24], does show up as an quite an important reference tool in [26]. Flattening is a useful operation to verify rank information for tensors, by converting them into matrices i.e. by lowering them from order-d tensors to order-2 tensors. Such an operation can also be used as a quick check when performing operations or constructions on tensors to detect changes in rank. This is due to the fact that computing the rank of a matrix is a much easier problem than computing the rank of an order d tensor, especially as d increases.

To define flattening, it is important to first specify how to *unravel* or *vectorise* a tensor of order d into a vector.

Definition 2.13 (Unravelling a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be a general order-d tensor. Then the vectorisation of T is denoted $T^{\emptyset}$ and is given by the following mapping:

$$\cdot^{\emptyset} : V_1 \otimes \cdots \otimes V_d \to V \cong \mathbb{F}^{\prod_{i=1}^d n_i}$$

taking

$$T = \sum_{k=1}^{r} v_1^k \otimes \cdots \otimes v_d^k \mapsto \sum_{k=1}^{r} \left(\sum_{J=(1,\ldots,1)}^{(n_1,\ldots,n_d)} \left(\prod_{i=1}^{d} v_i^k(J_i) \right) e_J \right)$$

Here e_J refers to the standard basis vectors of $\mathbb{F}^{\prod_{i=1}^d n_i}$, ordered in lexicographic fashion. Unravelling/vectorising is a well defined, linear operation.

Remark. Unravelling/vectorising tensors does depend on the choice of basis vectors made for each of the V_i.

Definition 2.14 (Flattening a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be a general, order-d tensor. Then

$$T := \sum_{k=1}^{r} v_1^k \otimes \cdots \otimes v_d^k,\ v_j^k \in V_j \ \forall\ k = 1, \ldots, r.$$

The flattening of T along the i-th mode is denoted $T^{(i)}$ and lies in the space $V \otimes V_i$, given by

$$T^{(i)} = \sum_{k=1}^{r} (v_{\hat{i}}^k)^{\emptyset} \otimes v_i^k.$$

Here $V \cong \mathbb{F}^{\prod_{j \neq i} n_j}$ containing all the unravelled tensors from $V_{\hat{i}}$.

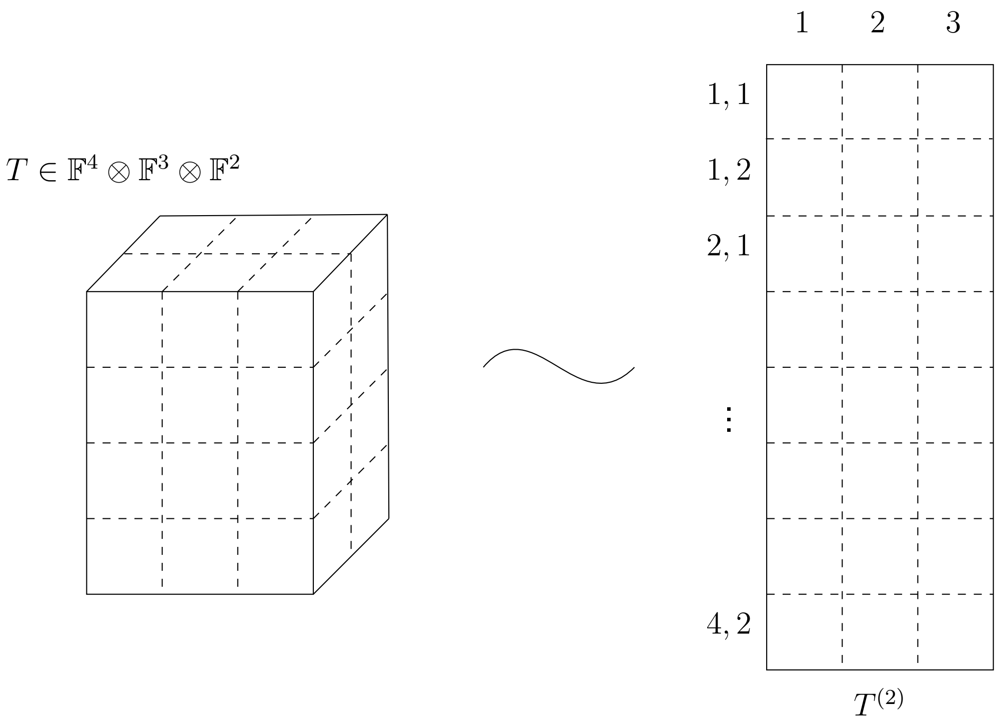

Figure 4: Flattening a tensor T

Example 2.5. Consider a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, where $T := (1,2,3,4) \otimes (1,2,3) \otimes (1,2)$. Then flattening the tensor along the 2-mode can be computed as

$$T^{(2)} = ((1,2,3,4) \otimes (1,2))^{\emptyset} \otimes (1,2,3).$$

While this looks strange, the vector required to be tensored with $(1,2,3)$ is formed by flattening the matrix $(1,2,3,4) \otimes (1,2)$ along the 1-mode. This is equivalent to the vector $(1,2,3,4,2,4,6,8)$. Figure 4 provides an illustration of this example, also highlighting how the column and row indices are generated.

Remark. Each column $T_j^{(i)} := T^{(i)}(-, j)$ represents the vectorisation of the j-th i slice of T, denoted $(T_j^i)^{\emptyset}$.

Much like how the slices of a tensor can be generated as the image of a slicing operator, the flattenings along the various modes of a tensor can be described by defining an operator.

Definition 2.15 (Flattening Operator)**.** Given the space $V_1 \otimes \cdots \otimes V_d$, and integers $1 \leq k \leq d$, define the operation

$$F^{(k)} : V_1 \otimes \cdots \otimes V_d \to V \otimes V_k, T \mapsto T^{(k)}.$$

$F^{(k)}$ is a well-defined, linear operator.

Remark. In the literature surveyed, flattening can be done not just along one mode, but along any collection $S \subseteq \{1, \ldots, d\}$. This explains the empty-set notation denoting the unravelling of a vector. Since the multiple-mode idea is not used in the constructions that follow in this thesis, this more general definition of flattening is not covered.

When dealing with symmetric tensors, flattening along any mode is equivalent, since the tensor is symmetric. Therefore $T^{(i)} = T^{(j)}$ for all $i \neq j$. The resulting matrix is an element of $\mathbb{F}^{(d-1)n} \otimes \mathbb{F}^n$.

2.5 The Geometry of Tensors

Definition 2.16 (Segre Mapping)**.**

Definition 2.17 (Veronese Mapping)**.**

Definition 2.18 (k-th Secant Variety)**.**

Chapter 3

Literature Review

Since the statement of the conjecture in 2008, there has been a substantial amount of work done to arrive at a conclusive final resolution to the problem. Some of this body of work relies on the specific geometric construction of secant varieties and how they relate to the question of computing rank and symmetric rank. This connection to geometry is not surprising because tensors can also be seen as points lying in some topological space (much like vectors and matrices can be seen as representing points, as well as algebraic objects). This chapter goes over some of the work on proving and disproving Comon, with the final section of this chapter compiling all the conditions currently known for which the conjecture holds or doesn't hold.

3.1 Attempts to Prove the Conjecture

The statement of Comon's Conjecture can now be stated in more formal terminology, as introduced in the previous chapter.

Conjecture (Comon et al.[12]). *Given a symmetric, order-d tensor $T \in S^d(\mathbb{F}^n) \subset (\mathbb{F}^n)^{\otimes d}$, then* ***Rk***$(T) =$ ***SRk***(T).

The ability to rewrite symmetric tensors of fixed order both as homogenous polynomials and as a sum of regular tensors was the key factor that seems to have inspired this conjecture, as stated in subsection 4.4 from Comon et al. [12]. This section details some of the papers that prove that the conjecture holds under certain constraints.

As mentioned in the introduction Comon et al. [12] showed that the conjecture was true for tensors of "large enough" order — a notion that became concrete in later literature. To establish such a proof, the authors first proved the following lemma (stated as a proposition), highlighting how a finite collection of linear polynomials, under pairwise independence constraints could be used to generate a linearly independent set of degree-d homogenous polynomials. Doing so requires the following definition:

Definition 3.1 (Bilinear form on the space of homogenous polynomials [12]). Let $F, G \in \mathbb{F}[x_1, \dots, x_n]_d$, where

$$F(\mathbf{x}) = \sum_{|\alpha|=d} \binom{d}{a_1, \dots, a_n} \lambda_\alpha \mathbf{x}^\alpha, \quad G(\mathbf{x}) = \sum_{|\alpha|=d} \binom{d}{a_1, \dots, a_n} \mu_\alpha \mathbf{x}^\alpha.$$

Here $\alpha = (a_1, \ldots, a_n)$ and $\lambda_\alpha, \mu_\alpha \in \mathbb{F}$ for all α. Then the form $\langle \cdot, \cdot \rangle : \mathbb{F}[x_1, \ldots, x_n]_d \times \mathbb{F}[x_1, \ldots, x_n]_d \to \mathbb{F}$ defined as

$$\langle F, G \rangle := \sum_{|\alpha|=d} \binom{d}{a_1, \ldots, a_n} \lambda_\alpha \mu_\alpha,$$

is a symmetric, non-degenerate bilinear form.

Lemma 3.1 ([12]). *Let $L_1, \ldots, L_r \in \mathbb{F}[x_1, \ldots, x_n]_1$, be linear forms in n variables. Suppose for all $i \neq j$, $L_i \neq \alpha L_j$ (where $\alpha \in \mathbb{F}$). Then for any $d \geq r-1$, the polynomials $L_1^d, \ldots, L_r^d$ are linearly independent in $\mathbb{F}[x_1, \ldots, x_n]$.*

Proof. Fix $d \geq r-1$. Suppose if possible, that there exist coefficients $\alpha_1, \ldots \alpha_r \in \mathbb{F}$ such that $\sum_{i=1}^r \alpha_i L_i^d = 0$. Since the bilinear form is non-degenerate, the expression

$$\langle F, \sum_{i=1}^r \alpha_i L_i^d \rangle = 0$$

for any $F \in \mathbb{F}[x_1, \ldots, x_n]_d$. This simplifies to

$$\sum_{i=1}^r \alpha_i \langle F, L_i^d \rangle = \sum_{i=1}^r \alpha_i F(L_i) = 0$$

where the expression $F(L_i) = F(\beta_1, \ldots, \beta_n)$ when $L_i = \sum_{i=1}^n \beta_i x_i$. The next step is to show that a particular degree-d homogenous polynomial F exists, such that it vanishes on all except one of the L_i. Consider F, a degree-d homogenous polynomial, to be the multiple of a product of $r-1$ linear forms, such that $F(L_r) \neq 0$ but $F(L_i) = 0$ for all $i < r$. Then one can determine that $\alpha_r = 0$, by the bilinear form expansion derived above. In a similar manner, similar manner, it can be shown that all the other α_i are also zero, leading to the conclusion that the forms L_i^d are linearly independent. □

This lemma also allows for determining when symmetric, order-d tensors can be considered as linearly independent, using the isomorphsim in Proposition 2.1.

Proposition 3.1 ([12]). *Let $v_1, \ldots, v_s \in \mathbb{F}^n$ be pairwise linearly independent. If d is sufficiently large, then the symmetric tensor, defined as*

$$T := \sum_{i=1}^s v_i^{\otimes d}$$

*satisfies **Rk**(T) = **SRk**(T) generically (i.e in a dense subset of order-d symmetric tensors).*

Proof. To begin with, since $S^d(\mathbb{C}^n) \subseteq (\mathbb{C}^n)^{\otimes d}$, for all $T \in S^d(\mathbb{C}^n)$, **Rk**$(T) \leq$ **SRk**(T). It is enough to show that in this specific case, for a tensor T as defined in the statement of this proposition, the reverse inequality i.e. **Rk**$(T) \geq$ **SRk**(T) holds.

Suppose **Rk**$(T) = r$ and **SRk**$(T) = s$. Then there exist decompositions

$$\sum_{i=1}^r x_1^i \otimes \cdots \otimes x_d^i = T = \sum_{j=1}^s v_j^{\otimes d}.$$

Without a loss of generality, assume that d is even. Then T can be written instead as $T = \sum_{i=1}^{s} v_j^{\otimes d/2} \otimes v_j^{\otimes d/2}$. By the previous lemma, for $d/2 \geq s-1$, the tensors $v_j^{\otimes d/2}$ are generically linearly independent. Now therefore, one can find functionals $\Phi_1, \ldots, \Phi_s \in S^{d/2}(\mathbb{F}^n)^*$ that are duals to the vectors $v_j^{\otimes d/2}$. i.e. $\Phi_i(v_j^{\otimes d/2}) = \delta_{i,j}$, the Kronecker delta. Then the initial equality of decompositions can be transformed (after contracting along the first $d/2$ modes) into the following expressions

$$\sum_{i=1}^{r} \alpha_{ij} x_{d/2+1}^i \otimes \cdots \otimes x_d^i = v_j^{\otimes d/2},$$

for each of the v_j, where $\alpha_{ij} = \Phi_j(x_1^i \otimes \cdots \otimes x_{d/2}^i)$ where $i = 1, \ldots, r$. This then implies that the s linearly indepenedent vectors $v_j^{\otimes d/2}$ lie in the span of the r tensors of the form $\{x_{d/2+1}^i \otimes \cdots \otimes x_d^i\}_{i=1}^r$. This then implies that $r \leq s$, and the proof is complete. $\square$

The authors further noted that, as a result of the Veronese embedding (a canonical construction in algebraic geometry), the rank of a symmetric tensor $T \in S^d(\mathbb{C}^n)$ cannot be more than $\binom{n+d-1}{d}$. Another major result in their paper was that the conjecture held generically whenever $\textbf{SRk}(T) \leq n$, the dimension of the underlying vector space. In particular, they were able to show that for every $T \in S^d(\mathbb{C}^n)$, the conjecture held if $\textbf{SRk}(T) = 1$ or 2.

A concurrent approach to tackling Comon's conjecture arises from the transposition of a classical problem from number theory — the Waring decomposition of a number — to the language of polynomials. This famous problem then gets restated as follows:

"What is the the minimum number of linear polynomials required such that a linear combination of them to the d-th power is equal to a homogenous degree-d polynomial?"

In mathematical notation, take $f \in \mathbb{F}[x_1, \ldots, x_n]_d$, a homogenous polynomial of degree d and $\mathbf{a} := (a_1, \ldots, a_n)$ denoting a vector containing coefficients such that $\mathbf{a} \cdot \mathbf{x} := a_1 x_1 + \cdots + a_n x_n$. Then the Waring problem for polynomials is the following minimisation problem

$$f := \sum_{|\alpha|=d} f_\alpha \mathbf{x}^\alpha = \min_{s>0} \sum_{i=1}^{s} (\mathbf{a}_i \cdot \mathbf{x})^d \tag{3.1}$$

A solution to the polynomial Waring problem solves the issue of computing symmetric rank for a given symmetric tensor (see Section 2.3). In fact this relationship is so significant that in the literature, the term **Waring Rank** is used synonymously with symmetric rank when discussing symmetric tensors. An equivalent formulation for the Waring problem is in terms of predicting the dimensions of higher secant varieties and relating these predictions to the symmetric (or Waring) decompositions for symmetric tensors [4].

The polynomial Waring problem for generic forms was solved by Alexander and Hirschowitz in 1995, by using this secant variety formulation, building on previous work from Campbell, Palatini and Terracini, and using the novel "differential Horace's method" [4]. They showed that $\sigma_r(\nu_d(\mathbb{P}\mathbb{F}^{n+1}))$ - the r-th secant variety to the d-th

Veronese embedding of $\mathbb{F}^{n+1}$ - is of expected dimension with 5 exceptional cases. Brambilla and Ottaviani [6] in 2008 gave a self-contained proof of the theorem, with some simplifications, particularly in the case of order 3 symmetric tensors. The existence of concrete methods for computing rank and symmetric rank meant that for any given symmetric tensor, Comon's conjecture could be quickly verified or disproved.

Ballico and Bernardi [2] in 2013 showed that that a tensor T lying on the tangent variety to a Veronese variety must always have **Rk**(T) = **SRk**(T). Their paper highlighted explicit algorithms for the computation of the rank of a tensor that could be expressed as the limit of a linear combination of elementary tensors. The main theorem in their paper is as follows. The Segre variety as a classical construction in algebraic geometry is used to visualise elementary tensors that live in a tensor product of vector spaces. The Veronese variety (again a classical construction), can be seen as embedded into this larger variety, showing only the elementary symmetric tensors. Thus general points in the spaces containing the Segre (or Veronese) can be seen as general tensors.

Theorem 3.1 ([2])**.** *Let $\tau(X_{n_1,\dots,n_d})$ be the tangential variety of the Segre variety $X_{n_1,\dots,n_d}$. Then, for each $P \in \tau(X_{n_1,\dots,n_d})$, one has that*

$$\textbf{\textit{Rk}}_{X_{n_1,\dots,n_d}} = \eta_{X_{n_1,\dots,n_d}}(P),$$

where $\eta_{X_{n_1,\dots,n_d}}(P)$ denotes the minimum size of a set $E \subseteq \{1,\dots,d\}$ such that P as a linear subspace lies in the linear span of exactly $|E|$ linear subspaces, passing through a point $O \in X_{n_1,\dots,n_d}$

Remark. This theorem equates the minimum number of points lying anywhere on the Segre variety whose linear span contains the point P lying in the tangential variety, and the minimum number of linear spaces generated at a point O on the Segre in whose linear span P must lie. The application to Comon's Conjecture comes from the following corollary

Corollary 3.1. *Suppose P lies in the linear span of any two points lying on the Veronese (embeddable inside the Segre). Then **Rk**(P) = **SRk**(P).*

Computing decompositions covers the first question about symmetric rank — the existence of a minimal length symmetric decomposition. The other question, concerning the uniqueness of a decomposition, requires the definition of when a tensor is *identifiabililty*.

Definition 3.2 (Identifiability)**.** A tensor $T \in V_1 \otimes \cdots \otimes V_d$ is said to be **identifiable** if it admits a unique rank decomposition (modulo scaling by d-th roots of unity and permutations of summands)

Example 3.1. The tensor $(3,4)\otimes(1,1)+(4,5)\otimes(1,0)$ can be written as $(7,9)\otimes(1,0)+(3,4)\otimes(0,1)$, and is thus not identifiable, since it can be decomposed in distinct ways, not involving simply a permutation of summands or a direct rescaling.

Generic identifiability is the idea that almost every tensor (i.e. a tensor in a dense open subset of tensors), is identifiable. Specific identifiability poses the opposite question - given a specific tensor T, does it admit a unique decomposition? Chiantini, Ottaviani and Vannieuwenhoven [10, 11] in two subsequent papers tackled both of these questions. Their earlier paper showed that a general symmetric tensor in $S^d(\mathbb{C}^{n+1})$ of sub-generic rank r was identifiable, provided $r \leq \binom{n+d}{d}/(n+1)$ (with all exceptions again being covered by the Alexander-Hirschowitz theorem). Their second paper focused on showing what criteria could be called "effective" to determine if a given tensor was specifically identifiable. Criteria were designated effective if their conditions were satisfied by generic points (i.e. the criteria were satisfied in an open, dense subset). One of the conclusions of this last paper was that Kruskal's criterion (a condition on collections of order-d tensors, seen as points in a vector space) was *effective*. Following Section 7 in [11], the authors address the applications of their results to Comon's conjecture via the following theorem

Theorem 3.2 ([11]). *Let*

$$T = \sum_{i=1}^{r} \phi_i v_i^{\otimes d},$$

with $\phi_i \in \mathbb{F} \setminus \{0\}$, and $v_i \in \mathbb{F}^n$. Then T is a generic, order-d, symmetric tensor, lying in $(\mathbb{F}^n)^{\otimes d}$. If

$$r \leq \begin{cases} \frac{3}{2}n - 1 & d = 3 \\ \binom{k+n}{n} + \frac{1}{2}\binom{2+n}{n} - 1 & d = 2k+1 \\ \binom{k+n}{n} - n - 1 & d = 2k, \end{cases}$$

with $2 \leq k \in \mathbb{N}$, then T admits only symmetric (or Waring) decompositions, and the symmetric and tensor rank of T must coincide

Friedland [14] in 2016 focused not on the computation of the symmetric rank, but on the properties of symmetric tensors of a given symmetric rank. In particular the paper focused on proving the following theorem, before moving to other versions of Comon's conjecture:

Theorem 3.3 ([14]). *Let $d \geq 3$, and $|\mathbb{F}| \geq 3$. Let $T \in S^d(\mathbb{F}^n)$. Suppose that **Rk**$(T) \leq$ **Rk**$(T^{(i)}) + 1$, for $1 \leq i \leq d$. Then **SRk**$(T) =$ **Rk**(T).*

The proof is divided up into smaller sections, and uses lemmas regarding the Kruskal rank to determine linear independence in a set of vectors, and therefore the rank of a generic order-d tensor formed using said vectors. The authors first address the case where **Rk**$(T) =$ **Rk**$(T^{(i)})$ and $d \geq 3$. They then move on to covering the case where $d = 3$, and **Rk**$(T) =$ **Rk**$(T^{(i)} + 1)$, before finally covering the cases where $d \geq 4$. Their paper also specifies that $|\mathbb{F}| \geq 3$ because for $\mathbb{F} = \mathbb{Z}_2$, it can be shown that not every order-3 symmetric tensor is a sum of elementary symmetric tensors (see [14], Section 4).

Zhang et al. [27] conclusively showed that Comon's conjecture held for all tensors where the rank of the tensor under consideration is not more than its order. The main theorem proved is as follows:

Theorem 3.4 ([27]). *Let $T \in S^d(\mathbb{F}^n)$ with $d \geq 3$. Then if **Rk**$(T) < d$, any rank decomposition of T is a symmetric rank decomposition.*

Two corollaries follow from this theorem in their paper, providing the positive answers to Comon's Conjecture.

Ballico et al [3] at the end of 2018, refocused their efforts on the question of identifiability, and worked on refining Kruskals criterion (sharp and algebraic in nature) into a geometric criterion that can be verified more easily. They concluded that Comon's conjecture held for tensors whose symmetric rank r is strictly bounded above by $\binom{n+e}{e}$ for a specific choice of e. This choice depended upon splitting the Segre product (another classical construction from algebraic geometry) into two disjoint components and then imposing conditions on a set of points on the Segre variety that exhibited distinct behaviour when projected down to the first component versus the second.

Theorem 3.5 ([3])**.** *Fix a partition $E \sqcup F = \{1, \ldots, d\}$ such that $E = \{a_1, \ldots, a_{d-c}\}$ and $F = b_1, \ldots, b_c$ for a fixed $0 < c < d$. Further consider Y to be the Segre variety corresponding to the d spaces - $\mathbb{PF}^{n_1}, \ldots, \mathbb{PF}^{n_d}$. Let M_F be the affine dimension of the space containing the Segre variety corresponding to only the F-spaces — $\mathbb{PF}^{n_{b_1}}, \ldots, \mathbb{PF}^{n_{b_c}}$. Thus $M_F := \prod_{i=1}^{c} n_{b_i}$. Fixing $0 < c < M_F$, consider $A \subset Y$ a zero-dimensional scheme such that the following two conditions are satisfied*

1. *$h^1(\mathcal{I}_A^E(1, \ldots, 1)) = 0$ - the difference between the degree of image of the scheme $\nu(A)$ in the E-factor Segre, and the affine dimension of the span.*
2. *$h^0(\mathcal{I}_A^F(1, \ldots, 1)) < M_F - c$ - the co-dimension of the linear span of the image of $\nu(A)$ in the F-factor Segre.*

Then, taking any $T \in \langle \nu(A) \rangle$, such that T doesn't lie in the linear span of the image of any subspace $A' \subsetneq A$, there must exist no 0-dimensional schemes $B \subset Y$ with $\deg(B) \leq c$ such that $T \in \langle \nu(B) \rangle$.

The full statement of this theorem involves working with schemes as they relate to tensors, but the application to Comon's conjecture comes from a corollary and remark following it, given as follows:

Corollary 3.2 ([3])**.** *Consider a zero-dimensional scheme $A \subset \mathbb{PF}^n$, of degree $c + 1$. Let $\mathcal{J}_A$ be the ideal sheaf of A in $\mathbb{PF}^n$ and assume that $h^1(\mathcal{J}_A(e)) = 0$, for some $e \leq d/2$. Taking $T \in \langle \nu_d(A) \rangle$, again such that it is not contained in the linear span of any subspace $A' \subsetneq A$, if A is reduced, then T as a tensor has **Rk**$(T) = c + 1$.*

Remark ([3])**.** Assuming $d = 2e$, implies Comon's conjecture holds for symmetric tensors T having a minimal symmetric decomposition A, with $h^1(\mathcal{J}_A(e)) = 0$.

The authors of this paper clearly pointed out that even though they had increased the scope of validity of Comon's conjecture, Shitov's proposed counterexample [21] lay outside this scope and thus might have still been a plausible counterexample.

Casarotti et al. [9] in 2018 proposed to improve the bounds for Comon's conjecture when working with flattenings, provided that equations for the secant varieties associated to the Veronese variety could be generated. Their result was for a tensor T having rank $r < \binom{n+\lfloor \frac{d}{2} \rfloor}{n}$, the **SRk**$(T) = r$, in cases where such generating determinantal equations were known.

Proposition 3.2 ([9]. Note here that this result is Casarotti et al. citing work by Iarobbino and Kanev.)**.** *For any integer $h < \binom{n+\lfloor \frac{d}{2} \rfloor}{n}$, there exists an open subset U_h in the h-secant variety to the Veronese variety $\nu_d(\mathbb{PF}^{n+1})$ such that for any $T \in U_h$, the rank and symmetric rank of T coincide.*

Finally in a paper from 2020, Seigal [20] demonstrated that for all ”cubic surfaces”, Comon’s conjecture does hold. Here cubic surfaces were used to refer to the fact that under the identification with homogenous polynomials, order-3 symmetric tensors correspond to cubic polynomials, and thus their zero sets could be called cubic surfaces. The author also used flattenings of tensors (see Section 2.4.2), dividing their proof into cases depending on the number of singularities on the cubic surfaces in question.

Theorem 3.6 ([20])**.** *The rank and symmetric rank agree for cubic surfaces i.e. symmetric $4 \times 4 \times 4$ tensors.*

Corollary 3.3 ([20])**.** *The rank and symmetric rank of a cubic polynomial in n variables are the same, whenever the symmetric rank is ≤ 7.*

3.2 Attempts to Disprove the Conjecture

In spite of all of these preliminary results indicating positive answers towards Comon’s Conjecture, Yaroslav Shitov seemed to arrive at a method of constructing a potential counterexample, as early as 2018 [21]. An error found in this paper led to the voiding of this specific counterexample in 2024, but the period of six years in the interim led to multiple other preprints from Shitov [22–24].

These preprints started with a newer, alternative methodology for the construction of counterexamples. While there is now no material carrying the stamp of peer-review and Shitov’s name, his initial contribution and subsequent preprints seem to indicate a viable approach to constructing a counterexample to Comon’s conjecture over $\mathbb{R}$, if not $\mathbb{C}$.

3.2.1 Shitov’s First Paper in 2018

The 2018 paper began with taking the ground field as $\mathbb{C}$, and fixing the notation to be used. The main objects of interest in the paper were order-3 tensors, and the introductory section of the paper highlighted the behaviour of such tensors under elementary transformations with respect to their constituent order-2 ”slices”. The construction of the counterexample began with the description of a 100×100 matrix, partitioned into 20 submatrices.

Each submatrix was then mapped to a corresponding 100×100 matrix, now with entries having the same indices as the submatrix set to 1 and all other entries as 0. Taking the $1-$ and $2-$ supports of each of these, and pre- and post-multiplying them with a column-shift operator with parameter a led to another set of matrices of the same size (still 100×100).

It is at this point that the paper made another specific construction. A set $\mathcal{B}$ was constructed consisting of matrices of distinct formats (viz. 200×200 and 400×400). While there is no issue in defining such a set, the paper then constructed ”the $\mathbb{C}$-linear span” of the elements of this set. Addition of quantities that do not live in the same dimensional vector space, without a specified or implied embedding map (at least from this author’s knowledge) presented a major problem with following this paper.

1	1	1	11	9	9	9	10	10	10
1	1	1	2	2	2	12	10	10	10
1	1	1	2	2	2	3	3	3	13
4	4	14	2	2	2	3	3	3	4
4	4	5	5	5	15	3	3	3	4
4	4	5	5	5	6	6	6	16	4
7	17	5	5	5	6	6	6	7	7
7	8	8	8	18	6	6	6	7	7
7	8	8	8	9	9	9	19	7	7
20	8	8	8	9	9	9	10	10	10

Figure 5: Observation 8 from [21], describing the 20 submatrices. Each number in the partition stands for a 10×10 block.

Later in the paper a tensor decomposition was described (Eq. 5.6 in [21].) This stated decomposition was the subject of the erratum published in 2024 [13]. The erratum indicated such a decomposition was true only in specific cases, but could not be said to hold in general unless proven. Proving the statement has been a non-trivial challenge, and since no answer has been arrived at, the current consensus is that the paper [21] does not lead to a valid counterexample.

While certain choices were made during this process that led to unanswered questions, the overall tools for later establishing a valid counterexample were first laid out in this paper. These included the two minor operations - slicing and flattening, and the three major operations - adjoining slices to tensors, generating linear families modulo slices and cloning.

3.2.2 Shitov's Subsequent Preprints

In the following years, the preprints [22–24] from Shitov either tackled Comon's conjecture in a restricted setting, or developed tools to construct more general counterexamples. These might also be used to construct a counterexample to Strassen's conjecture (another tensor rank conjecture concerning the additivity of rank). The preprint from 2020 [22] tackled the conjecture over $\mathbb{R}$ and it was this preprint that formed the basis for the paper from Wang and Seigal [26].

The constructions made in each preprint followed the same pattern, indicating a common toolset and structure of approach. Each approach saw an order-d tensor as composed of lower order-$(d-1)$ tensors, which allowed for tensors to be transformed by "slice" along a specified mode. Viewing tensors in this manner opened a way to describe tensors modulo these slices — a useful operation. This kind of operation can be seen seen in both number theory — when working with integers modulo primes, and in algebraic geometry as a tool to study problems in a more localised setting. Such a family, generated by slices, gave an idea of the general properties of the orginal tensor (how many linearly independent slices along a specified mode, maximum slice rank along each mode etc.) Flattenings assisted with the computation of rank for these slices (because computing the rank of matrices is a known and easier problem to solve than doing so for higher order tensors) and the flattenings lent themselves well to describing rank information that might change in the relationship between a tensor and its slices.

These tools allowed for an analysis of any particular tensor, but the construction aspect of this approach became evident with the adjoining of slices to a tensor and the cloning operation. The former operation changed the format of the original tensor with respect to certain chosen slices, and allowed the rank information of the family generated modulo the slices to be preserved in a new tensor containing both the original and the slices that were required.

Cloning was a way to maintain rank information by simply making a higher format copy, and allowed for more freedom when modifying slices — simply because there were more to chose from. Cloning started to play a role because perturbing such an adjoined tensor (usually by the modification of one of the constituent slices) caused a violation of the properties that were supposed to be preserved.

While each of these preprints started with this general structure, and [24] proposed the construction of a general counterexample over any $\mathbb{F}$ with char$(\mathbb{F}) \neq 3$, an intermediate step involved in all of these was the construction of an *emulating family* of larger-format tensors. Such a family, if it existed, would allow for the rest of the constructions in each preprint to proceed. However, proving the existence of such a family was still not trivial. The latest preprint attempted to do so, highlighting a similar construction from [23], but tracing the construction back led to the definition of a function dependent on an arbitrary constant - in the same way as the paper [21] relied on the existence of an arbitrarily partitioned 100×100 matrix. Such choices, while perhaps valid, did not appear to be explained at any point in their corresponding documents, raising some skepticism and prompting further investigation into better documented methods that could explain them, or proofs to deny their existence.

3.3 Overview of Results related to the Conjecture

The current overview of cases where Comon's Conjecture is known to hold is as follows:

- For $T = \sum_{i=1}^{s} y_i^{\otimes d}$ and d *large enough* [12]. The large enough argument in the paper hinges on a classical result from algebraic geometry, that d points in $\mathbb{C}^n$ form the solution set of of polynomial equations of degree $\leq d$.

- For $T \in S^d(\mathbb{C}^n)$ and **SRk**$(T) = 1, 2$ [12].

- For $T \in S^d\mathbb{F}^n$, and **Rk**$(T) \leq \frac{dn-d+1}{2}$ where $\mathbb{F} = \mathbb{R}, \mathbb{C}$ [18].

- For $T \in S^d(\mathbb{C}^n)$ for $(d, n) \in \{(3, 2), (4, 2), (3, 3)\}$ [14].

- In the case $d \geq 3$, char$(\mathbb{F}) \geq 3, T \in S^d(\mathbb{F}^n)$ and **Rk**$(T) =$ **Rk**$(T^{(i)})$ or **Rk**$(T^{(i)}) + 1$ [14]

- When $T \in S^3(\mathbb{C}^n)$ and **Rk**$(T) \leq 5$ or **SRk**$(T) \leq 6$ [14].
- For $T \in S^d(\mathbb{F}^n)$ where **Rk**$(T) \leq d$ where $\mathbb{F} = \mathbb{R}, \mathbb{C}$[27].
- If T is a tensor lying on a tangential variety to a Veronese variety [2].
- If $T \in \mathbb{C}^2 \otimes \mathbb{C}^n \otimes \mathbb{C}^n$ [7].
- For T being a Coppersmith-Winograd tensor [16].
- Whenever a tensor T is cubic surface (real or complex) i.e. T can be represented by a homogenous cubic polynomial $\sum_{|\alpha|=3} \lambda_\alpha \mathbf{x}^\alpha$ [20].

The current list of counterexamples to Comon's conjecture is as follows:

- $T \in (\mathbb{R}^{5180})^{\otimes 6}$ having **Rk**$(T) = 30913 <$ **SRk**$(T) = 30914$ [26]

Chapter 4

Constructions Required for a Counterexample

This chapter mainly focuses on highlighting the newer constructions from Shitov's preprints [22–24], and concludes with an analysis of Wang and Seigal's work in [26] where they construct an $\mathbb{R}$-valued, order-6 symmetric tensor as a counterexample to the conjecture.

4.1 Major New Constructions from Shitov

While the previously defined operations of slicing and flattening are relatively "minor" in the grand scheme of Shitov's proposed counterexample to Comon, the following three major operations on tensors are integral to the final construction proposed (and indeed implemented) by Wang and Seigal [26].

In each of the following constructions, a number of order-$(d-1)$ tensors are used to modify an order-d tensor. Since this number is always finite, it makes sense to represent the lower-order tensors as i-slices of an order-d parametrising tensor (for a chosen $1 \leq i \leq d$). This makes defining the operations of adjoining and genererating a family modulo slices rely only on two order-d tensors, instead of an unspecified finite number of smaller slices.

Definition 4.1 (Creating a Tensor from a Family of Slices)**.** Let $d > 0$ and $\mathcal{M} = \{M_1, \ldots, M_N\} \in V_{\hat{i}}$ be a finite collection of order-$(d-1)$ tensors. Then each M_j can be written in the following manner

$$M_j := \sum_{k=1}^{r_j} m_{j,1}^k \otimes \cdots \otimes m_{j,i-1}^k \otimes m_{j,i+1}^k \otimes \cdots \otimes m_{j,d}^k.$$

Then the operation

$$\bar{\cdot}^i : \times_{i=1}^N V_{\hat{i}} \to V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots \otimes V_d,$$

defined by taking (as a tuple)

$$\mathcal{M} \mapsto \sum_{j=1}^N \sum_{k=1}^{r_j} m_{j,1}^k \otimes \cdots \otimes m_{j,i-1}^k \otimes e_j \otimes m_{j,i+1}^k \otimes \cdots \otimes m_{j,d}^k,$$

is a well-defined, bijective operation.

Remark. The order-d tensor so created, $\bar{\mathcal{M}}^i$ changes depending on how the slices M_j are arranged. One can denote the space this resulting tensor lies in as $\bar{V}_i(N)$. The N indicates the number of slices that were required. Since $\bar{\mathcal{M}}^i$ changes depending on how the slices are arranged, it is not possible to give a definition that is coordinate independent.

4.1.1 Background to Adjoining

Once the notion of a tensor of order d being composed of various order-$(d-1)$ slices has been established, one natural operation that can be performed is the *adjoininig* of slices along the d modes of the tensor, provided that each slice is of appropriate format. This caveat of appropriate format arises when trying to adjoin slices along multiple modes, because adjoining along any mode does result in a tensor with a different format from the original tensor. To perform this operation, one needs two pieces of information — the tensor to be transformed, and the collections of slices that are to be adjoined. This operation seems to have been first formally defined by Shitov in his 2018 paper [21], and further developed into the language below in his preprint from 2020[22].

In this thesis, adjoining is first defined along only one mode. To describe the adjoining process of collections along multiple modes, it is also necessary to describe how order-$(d-1)$ tensors can be padded so that they match the same format as the i-th slices of the tensor they are being adjoined to.

Definition 4.2 (Adjoining Slices to a Tensor along a specific Mode)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be a tensor, with format $(n_1, \ldots, n_d)$. Let $\mathcal{M} = \{M_1, \ldots, M_N\}$ be a collection of order-$(d-1)$ tensors lying in $V_{\hat{i}}$. Then define the following set

$$\tau := \{S^i_j(T)\}_{j=1}^{n_i} \sqcup \mathcal{M}.$$

Without a loss of generality, this set can be fit into a tuple of length $n_i + N$, with the first n_i entries describing the i-slices of T in order, and the last N entries the elements of $\mathcal{M}$ in order.

Then the operation of adjoining, $Adj^i : \times_{k=1}^{n_i+N} V_{\hat{i}} \to \bar{V}_i(n_i + N)$ is given by taking

$$\tau \mapsto \bar{\tau}^i.$$

Remark. Since the above definition relies on a coordinate dependent operation, it is similarly not possible to describe such an operation in a coordinate independent manner.

Post the adjoining operation, $Adj^i(\{S^i_j(T)\}_{j=1}^{P} n_i \sqcup \mathcal{M})$ lies in a tensor product of vector spaces over $\mathbb{F}$, but does not have the format $(n_1, \ldots, n_d)$. Instead it is a tensor now of format $(n_1, \ldots, n_{i-1}, n_i + N, n_{i+1}, \ldots, n_d)$.

Remark. The above definition allows for obtaining a tensor when adjoining a family of slices along only one mode. The next major step to be taken is adjoining slices along multiple directions. Doing so requires the definition of a *padding* operation, that allows for subsequent order-$(d-1)$ tensors to be adjusted into appropriate format, depending on the original order-d tensor, before being adjoined in the required direction.

Definition 4.3 (Padding an order-$(d-1)$ tensor without increasing order)**.** Let $d > 0$ be given, and $T \in V_1 \otimes \cdots \otimes V_d$ be the candidate tensor of format $(n_1, \ldots, n_d)$. It is according to this format that the given $A \in W_{\hat{i}}$ is to be padded. Now

$$A = \sum_{k=1}^{r} w_1^k \otimes \cdots \otimes w_{i-1}^k \otimes w_{i+1}^k \otimes \cdots \otimes w_d^k, w_j^k \in W_j \ \forall j = 1, \ldots, d.$$

Here $W_j \leq V_j$ for all $1 \leq i \leq d$, i.e. they are all subspaces of the V_j. A therefore has a format of $(m_1, \ldots, m_{i-1}, m_{i+1}, \ldots, m_d)$ where $m_j \leq n_j$. There exists a canonical injection $I_j : W_j \hookrightarrow V_j$ for all $j = 1, \ldots, d$, that appends 0's to each vector under consideration. Define the padding operation as

$$pad : W_{\hat{i}} \times V \to V_{\hat{i}}$$

taking

$$(A, T) \mapsto \sum_{k=1}^{r} I_1(w_1^k) \otimes \cdots \otimes I_{i-1}(w_{i-1}^k) \otimes I_{i+1}(w_{i+1}^k) \otimes \cdots \otimes I_d(w_d^k).$$

Example 4.1. Consider $T := (1,2,3,4) \otimes (5,6,7) \otimes (8,9) \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$ and $A := (1,1) \otimes (2,2) \in (\mathbb{F}^2)^{\otimes 2}$. Then $pad^1(A,T) = (1,1,0) \otimes (2,2) \in (\mathbb{F}^3)^{\otimes 2}$, the same format as the 1 slices of T. Likewise $pad^2(A,T) = (1,1,0,0) \otimes (2,2)$ and $pad^3(A,T) = (1,1,0,0) \otimes (2,2,0)$.

4.1.2 Adjoining Slices to a Tensor

The following section describes an algorithm for repeatedly applying the previous two definitions to construct a final tensor containing both the original smaller order-d tensor and all the slices along each of the d modes.

Algorithm: Adjoining order-$(d-1)$ slices to an order-d tensor along multiple modes.
Input: $d > 0$. A tensor $T \in W_1 \otimes \cdots \otimes W_d$ with format $(m_1, \ldots, m_d)$. Let $\mathcal{M}_i \subset V_{\hat{i}}$ be finite collections of order-$(d-1)$ tensors, each containing N_i slices.

Output: $Adj(T, \mathcal{M}_1, \ldots, \mathcal{M}_d)$

1. Let $i = 1$. Let $\tau^0 = T$.
2. Define $\tau^i := Adj^i(\{S_i^j(\tau^{i-1})\}_{j=1}^{n_i} \sqcup \mathcal{M}_i)$. Pass to Step (3)
3. Redefine $\mathcal{M}_j := \{pad(M_j^k, \tau_i)\}_{k=1}^{N_j}$ for each $j > i$. Pass to Step (4)
4. If $i \leq d$, set $i = i + 1$ and pass to step (2)

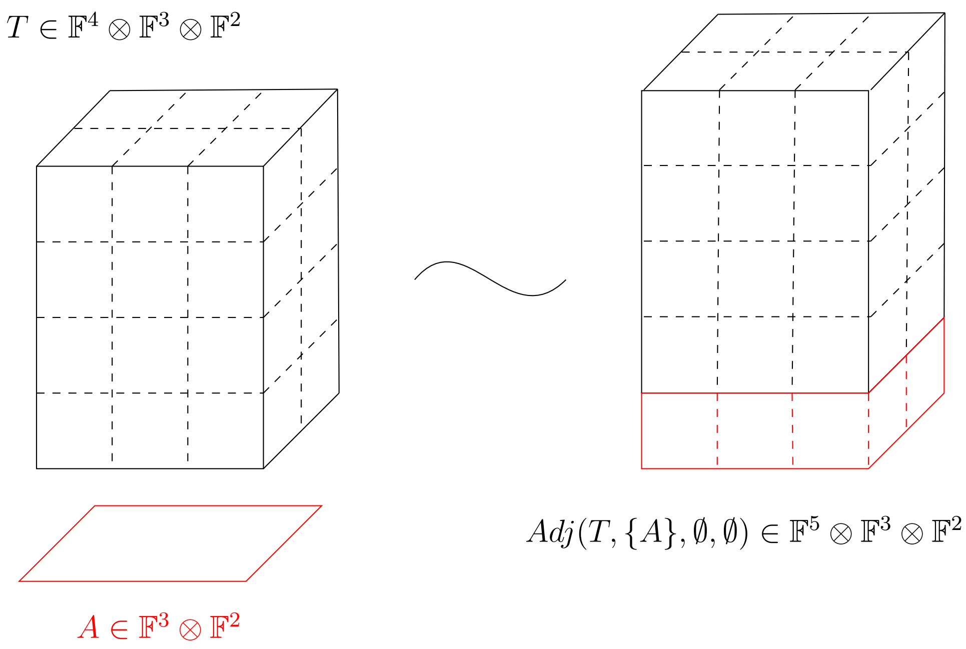

Figure 6: Adjoining a slice A to a tensor T

5. Return τ_d

The above algorithm defines the operation $Adj : W_1 \otimes \cdots \otimes W_d \times \left(\times_{i=1}^d \bar{W}_j(N_j)\right) \to V_1 \otimes \cdots \otimes V_d$.

Remark. The algorithm written above can be modified to work for any ordering of modes. Let $\sigma \in S_d$ be a permutation of $\{1, \ldots, d\}$, describing the new order of adjoining. The following steps must then be changed to work with this modified ordering:

- Step (2) changes to "Define $\tau^i := Adj^{\sigma(i)}(\{S^j_{\sigma(i)}(T)\}_{j=1}^{n_{\sigma(i)}} \sqcup \mathcal{M}_{\sigma(i)})$. Pass to Step (3)"
- Step (3) changes to "Redefine $\mathcal{M}_j := \{pad(M_j^k, \tau_i)\}_{k=1}^{N_j}$ for each $j \notin \{\sigma(k)\}_{k=1}^i$. Pass to Step (4)"

The output of this modified algorithm is the same as the original algorithm.

Observation. Figure 6 illustrates the above procedure for the case when only one slice is being adjoined in one direction. According to the above definition τ can be seen as an order-d tensor still, but now lying in $(V_1 \oplus \langle e_{1,n_1+1} \rangle) \otimes \cdots \otimes V_d$ where e_{1,n_1+1} indicates a "new" basis vector added onto the original vector space V_1.

The **symmetric adjoin** can be denoted $SAdj(T, \mathcal{M})$ where $\mathcal{M}$ is adjoined along every mode, following the same process as given in Section 4.1.2. Here $SAdj(T, \mathcal{M}) := Adj(T, \mathcal{M}, \ldots, \mathcal{M})$. The collection $\mathcal{M}$ consists entirely of symmetric order-$(d-1)$ tensors as well.

The adjoining process as described creates a tensor of larger format, but the rank of such a tensor may be appropriately constrained depending on the slices being adjoined. It is this constraining property that Shitov [24] and Wang and Seigal [26] use to construct an intermediate tensor with adjoined slices that has a differing rank and symmetric rank.

Remark. It may be trivial to note that while the order of the slices in $\mathcal{M}_i$ matters when constructing the final tensor, the rank information of the constructed tensor is independent of such changes in ordering. Further, the adjoining procedure itself can be modified, to follow any ordering of modes, as long as the slices required have been padded appropriately.

4.1.3 Generating Tensor Families modulo Slices

Given that a tensor can admit ny non-unique decompositions, one way to analyse the structure of a tensor is via an analogy to Gaussian elimination for matrices. The substitution method as described in [1, 16] has been a computational tool for analysing tensors since the 1960's. Rather than arriving at one possible "lowest" optimal tensor, generating a family of tensors modulo a set of slices is equivalent to looking at all the possible ways a tensor could be transformed, based on transforming its constituent slices along different modes.

Defining such an operation that generates a linear span necessitates one additional, alternate description of padding an order-$(d-1)$ tensor into a higher order tensor of appropriate format, now in terms of seeing the lower-order slice as embedded into an empty higher-order tensor. In doing so, it is also necessary to take a candidate higher order tensor as a "template" that the slice can be made to resemble.

Definition 4.4 (Padding an order-$(d-1)$ tensor into higher order)**.** Let $d > 0$, and $T \in V_1 \otimes \cdots \otimes V_d$, having a format $(n_1, \ldots, n_d)$. Let $A \in V_{\hat{i}}$ be an order-$(d-1)$ tensor, of the format $(n_1, \ldots, n_{i-1}, n_{i+1}, \ldots n_d)$, be of the following form:

$$A := \sum_{k=1}^{r} v_1^k \otimes \cdots \otimes v_{i-1}^k \otimes v_{i+1}^k \otimes \cdots \otimes v_d^k, v_j^k \in V_j \;\forall\; j = 1, \ldots, i-1, i+1, \ldots, d.$$

Since $\bar{V}_i(n_i) \cong V$, let Φ be an isomorphism. Then $Pad_j^i : V_{\hat{i}} \times V \to V$ can be defined as taking

$$(A, T) \mapsto \Phi \left(\sum_{k=1}^{r} v_1^k \otimes v_{i-1}^k \otimes e_j \otimes v_{i+1}^k \otimes \cdots \otimes v_d^k \right)$$

where e_j indicates the j-th basis vector for $\mathbb{F}^{n_i}$.

Remark. This definition still depends on the standard-ordered basis vectors for $\mathbb{F}^{n_i}$, and therefore cannot be made completely coordinate-independent.

Equipped with this method of padding one can now define slicing a general tensor T along one particular mode.

Definition 4.5 (Tensor Family Generated by Slices along a Mode)**.** Let $d > 0$ and $1 \leq i \leq d$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order-d tensor, and $\mathcal{M} := \{M_1, \ldots, M_N\} \subset V_{\hat{i}}$ be a finite collection of order-$(d-1)$ slices of the appropriate format.

Define $Mod^i(T, \bar{\mathcal{M}}) : V \times \bar{V}_i(N) \to 2^V$ in terms of coordinates as follows (2^P denoting the power set of a set P):

$$Mod^i(T,\bar{\mathcal{M}}) = \left\{ T + \sum_{k=1}^{n_i} \sum_{j=1}^{N} \lambda_{j,k} Pad_k^i(S_j^i(\bar{\mathcal{M}})) \mid \lambda_{j,k} \in \mathbb{F} \right\}.$$

Here slices are first selected from the constructed order-d tensor from the family of lower-order tensors, then these are padded appropriately to remain of the same format as T, and then scaled and added to T.

This definition can now be extended to performing this operation along all the d modes. The only difference now notationally is writing $Mod^j(Mod^i(T,\bar{\mathcal{M}}_i),\bar{\mathcal{M}}_j)$ to denote the $Mod^j(S,\bar{\mathcal{M}}_j)$ for every $S \in Mod^i(T,\mathcal{M}_i)$ and then taking the union of all the resulting sets. Thus the final family $F := T \bmod (\mathcal{M}_1, \ldots \mathcal{M}_d)$ is a result of the d-fold composition of the distinct Mod^i operations.

Proposition 4.1. *Let T be an order-d tensor $\in V_1 \otimes \cdots \otimes V_d$. Consider $\mathcal{M}_i \subset V_{\hat{i}}$ and $\mathcal{M}_j \subset V_{\hat{j}}$ as collections of lower-order tensors of format appropriate to the i and j slices of T respectively. Then the family generated modulo $\mathcal{M}_i$ and $\mathcal{M}_j$ is the same, regardless of which modulo operation is performed first, i.e.*

$$Mod^j(Mod^i(T,\bar{\mathcal{M}}_i),\bar{\mathcal{M}}_j) = Mod^i(Mod^j(T,\bar{\mathcal{M}}_j),\bar{\mathcal{M}}_i)$$

Proof.

$$\begin{aligned}
& Mod^j(Mod^i(T,\bar{\mathcal{M}}_i),\bar{\mathcal{M}}_j) \\
= \; & \left\{Mod^j(S,\bar{\mathcal{M}}_j) | S \in Mod^i(T,\bar{\mathcal{M}}_i)\right\} \\
= \; & \left\{ S + \sum_{a=1}^{n_j} \sum_{b=1}^{N_j} \lambda_{a,b} Pad_a^j(S_b^j(\bar{\mathcal{M}}_j)) | \lambda_{a,b} \in \mathbb{F}, S \in Mod^i(T,\bar{\mathcal{M}}_i) \right\} \\
= \; & \left\{ T + \sum_{c=1}^{n_i} \sum_{d=1}^{N_i} \lambda_{c,d} Pad_c^i(S_d^i(\bar{\mathcal{M}}_i)) + \sum_{b=1}^{n_j} \sum_{a=1}^{N_j} \lambda_{a,b} Pad_a^j(S_b^j(\bar{\mathcal{M}}_j)) | \lambda_{a,b}, \lambda_{c,d} \in \mathbb{F} \right\} \\
= \; & \left\{ S + \sum_{c=1}^{n_i} \sum_{d=1}^{N_i} \lambda_{c,d} Pad_c^i(S_d^i(\bar{\mathcal{M}}_i)) | \lambda_{c,d} \in \mathbb{F}, S \in Mod^j(T,\bar{\mathcal{M}}_j) \right\} \\
= \; & \left\{Mod^i(S,\bar{\mathcal{M}}_i) | S \in Mod^j(T,\bar{\mathcal{M}}_j)\right\} \\
= \; & Mod^i(Mod^j(T,\bar{\mathcal{M}}_j),\bar{\mathcal{M}}_i)
\end{aligned}$$

□

The above proof shows that the order of applying modulo operations on a given tensor does not affect the final resultant linear span.

Observation. Here, the family $T \bmod (\mathcal{M}_1, \ldots, \mathcal{M}_d) \subseteq V_1 \otimes \cdots \otimes V_d$, i.e. all its elements remain of the same order as the original tensor T. Figure 7 illustrates how one one such element from such a family can be generated.

Remark. As with the adjoining operation for symmetric tensors, generating a family of symmetric tensors modulo symmetric slices also follows immediately from the definitions provided above, but in this case it is the same family of order-$(d-1)$ slices, $\mathcal{M}$, being used along every mode. Therefore, for a tensor $T \in S^d(V)$, and $\mathcal{M} \subseteq S^{d-1}(V)$, one has $T \bmod \mathcal{M} := T \bmod (\mathcal{M}, \ldots, \mathcal{M})$.

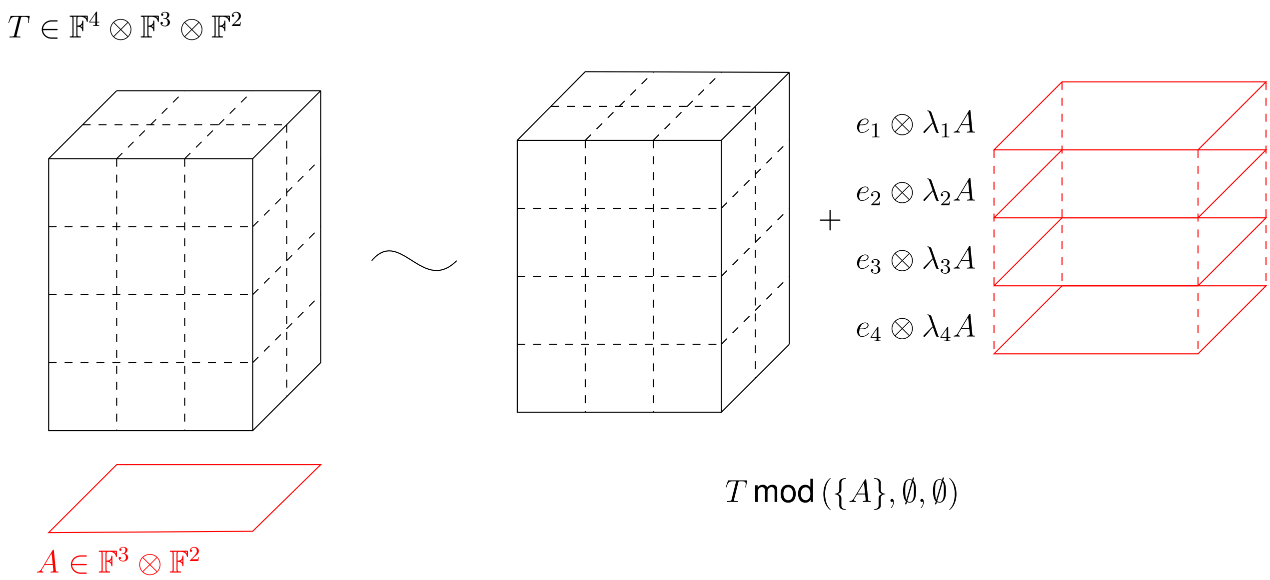

Figure 7: Family generated from a tensor T by a slice A

Since performing the modulo operation on a tensor gives a family instead of a single output tensor, one can only talk about how the maximal or minimal rank of the tensor *family* is affected. This is analogous to finding a minimal generating set/maximally linearly-independent set for a hyperplane in a vector space.

Combining Adjoining and Modulo

As seen in the previous two subsections, rank information of a tensor does change when adjoining slices, and the following theorem is becomes important in Wang and Seigal's 2023 paper [20]. In their paper, it is stated as a corollary to the real substitution method as defined in Lemma B.1 (also known as the layer reduction method) from [1].

Theorem 4.4 in [20] and Proposition 3.1 in [16] follow [1] in providing a proof for showing how the rank of a tensor changes when slices are modified along only one mode.

Lemma 4.1 (The substitution method [16,20,22,26]). *Let $T \in V_1 \otimes \cdots \otimes V_d$ be an order-d tensor. T can be written in terms of basis elements of V_1 in the following manner*

$$T = \sum_{k=1}^{n_1} e_k \otimes M_k, M_k \in V_{\hat{1}},$$

where $\{e_k\}_{k=1}^{n_1}$ forms a basis for V_1. Suppose $M_1 \neq 0$. Then, there exist constants $\lambda_2, \ldots, \lambda_{n_1}$ such that the tensor

$$\tau := \sum_{k=2}^{n_1} e_k \otimes (M_k - \lambda_k M_1)$$

*exists having **Rk**$(\tau) \leq$ **Rk**$(T) - 1$. Equality holds when M_1 is an elementary tensor.*

Proof. Suppose T has rank r. Then $T := \sum_{i=1}^{r} T_i$. Each of these elementary T_i can be further expressed in terms of the basis vectors for V_1 in the following manner $T_i = \sum_{k=1}^{n_1} a_{i,k} e_k \otimes L_k$, where each L_k is an elementary tensor in $V_{\hat{1}}$. Thus each 1-slice of T, given by M_k can be written as $M_k = \sum_{i=1}^{r} a_{i,k} L_k$. Thus in particular, since $M_1 \neq 0$, there is at least one $a_{i,1}$ is non-zero. Without a loss of generality, assume $a_{1,1} \neq 0$. Then setting each of the $\lambda_k = a_{1,k}$ the tensor

$$\tau := \sum_{k=2}^{n_1} e_k \otimes (M_k - \lambda_k M_1)$$

has all its slices $M_k - \lambda_k M_1$ expressible as a linear combination of the elementary $L_2, \dots, L_k$. Therefore, **Rk**$(\tau) \leq r - 1$. Now if M_1 was elementary, subtracting multiples of it from T only changes the rank by at most 1. □

Such coefficients can be found along any mode, for any finite collection of slices (not necessarily the ones that make up the original tensor, see Corollary B.2 in [1]). Thus one has the following inequality

$$\min \textbf{Rk}(T \,\text{mod}\, (\mathcal{M}_1, \dots, \mathcal{M}_d)) \leq \textbf{Rk}(T) - \sum_{i=1}^{d} \dim\langle \mathcal{M}_i \rangle.$$

Lemma 4.2. *For any tensor $T \in V_1 \otimes \cdots \otimes V_d$, the following statement always holds*

$$\min \textbf{\textit{Rk}}(T \,\textit{mod}\, (\{S^1_{j_1}(T)\}_{j_1=1}^{n_1}, \dots, \{S^d_{j_d}(T)\}_{j_d=1}^{n_d})) = 0$$

Proof. By the previous lemma, modifying a tensor T by a slice it contains $S^i_j(T)$ lowers the rank of the tensor by at least 1. Modifying T by all such slices $\{S^i_j(T)\}_{j=1}^{n_i}$ lowers the rank by $n_i - \dim\langle S^i_j\rangle_{j=1}^{n_i}$. For at least one mode i, the number of linearly independent slices is exactly the rank of the tensor (Lemma 3.1). This leads to the 0-tensor lying in the family $T \,\text{mod}\, (\{S^1_{j_1}(T)\}_{j_1=1}^{n_1}, \dots, \{S^d_{j_d}(T)\}_{j_d=1}^{n_d})$. □

Theorem 4.1 ([26]). *Suppose $T \in V := V_1 \otimes \cdots \otimes V_d$ is an order-d tensor, and $\mathcal{M}_i \subset V_{\hat{i}}$ for every $1 \leq i \leq d$. Then*

$$\textbf{\textit{Rk}}(\textit{Adj}(T, \mathcal{M}_1, \dots, \mathcal{M}_d)) \geq \min \textbf{\textit{Rk}}(T \,\textit{mod}\, (\mathcal{M}_1, \dots, \mathcal{M}_d)) + \sum_{j=1}^{d} \dim\langle \mathcal{M}_j \rangle$$

Equality holds if each of the $\mathcal{M}_i$ consist of only rank-one tensors.

Proof. Suppose T is an order-d tensor, having **Rk**$(T) = r$, and each $\mathcal{M}_i := \{M_{i,k}\}_{k=1}^{N_i}$. Then the order-$d$ tensor created by adjoining all of these slices along the appropriate modes is given as

$$\tau := \textit{Adj}(T, \mathcal{M}_1, \dots, \mathcal{M}_d).$$

The goal of this proof is to show that the adjoining of slices along all modes can be characterised in terms of the way these d distinct slice families interact with the original tensor T. Consider the following relationship seen as a consequence of the LEMMA FROM CHAPTER 2**Rk**$(\tau) = \max_{1 \leq i \leq d} \dim \mathcal{L}_i(\tau)$. Here

$$\mathcal{L}_i(\tau) := \textit{span}\left(\{S^i_j(\tau)\}_{j=1}^{n_i} \sqcup \{\textit{pad}^i(M_{i,k}, \tau)\}_{k=1}^{N_i}\right),$$

where first the i-slices containing the information from T are selected from τ, and then the appropriately padded slices containing new information. However, the non-trivial parts of the slices $pad^i(M_{i,k}, \tau)$ only interact with the $S^i_j(T)$ parts of the slices $S^i_j(\tau)$, and so it is seen that

$$\dim \mathcal{L}_i(\tau) = \dim span(\{S^i_j(T)\}_{j=1}^{n_i} \sqcup \{M_{i,k}\}_{k=1}^{N_i}).$$

By Lemma 4.1, this can be seen as the upper bound of

$$\min \textbf{Rk}(T \operatorname{mod} (\emptyset, \dots, \emptyset, \mathcal{M}_i, \emptyset, \dots, \emptyset)) + \dim\langle \mathcal{M}_i \rangle.$$

Therefore the following inequality exists where

$$\textbf{Rk}(\tau) \geq \max_{1 \leq i \leq d} \min \textbf{Rk}(T \operatorname{mod} (\emptyset, \dots, \emptyset, \mathcal{M}_i, \emptyset, \dots, \emptyset)) + \dim\langle \mathcal{M}_i \rangle.$$

This inequality describes the relationship between the adjoined construction, and the original when altered only along one mode. Such an inequality persists over alterations over any collection of modes. This is because the original characterisation of $\textbf{Rk}(\tau)$ in terms of the slice spaces could also have been done in terms of the ranks of all possible order-2 tensors contained within, and even in terms of all possible order-d tensors. Thus one can replace the $\max_{1 \leq i \leq d} \min \textbf{Rk}(T \operatorname{mod} (\emptyset, \dots, \emptyset, \mathcal{M}_i, \emptyset, \dots, \emptyset)) + \dim\langle \mathcal{M}_i \rangle$ term with

$$\min \textbf{Rk}(T \operatorname{mod} (\mathcal{M}_1, \dots, \mathcal{M}_d)) + \sum_{i=1}^{d} \dim\langle \mathcal{M}_i \rangle.$$

The equality portion of this relationship when all the $\mathcal{M}_i$ are made up of elementary tensors holds, again because Lemma 4.1 shows equality when elementary tensors are used. □

4.1.4 Cloning Tensors

The idea of cloning a tensor is born out of necessity. When constructing an emulating family for the proposed order-4 counterexample (see Section 4.2.2), difficulties arise when working with tensors of the specified format. Cloning a tensor, as the name suggests, seeks to preserve the properties of the original, while embedding it in a much "larger space".

Definition 4.6 (Cloning a Tensor)**.** Fix $d > 0$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order-d tensor.

The n-th cloning function $C_n : V \to (\bigoplus_{i=1}^n V_1) \otimes \cdots \otimes (\bigoplus_{i=1}^n V_d)$ can be defined in coordinates in the following manner:

$$C_n(T)(k_{1,j_1}, \dots, k_{d,j_d}) = T(k_1, \dots, k_d).$$

Here, the notation $k_{i,j_i} := k_{(j_i-1)n+i}$. where $1 \leq j_i \leq n$.

Observation. Since the definition of cloning depends on a non-zero d, one can clone tensors of any order, in particular vectors, as well. This leads to the description of the clone of a standard ordered basis vector $e_i \in V$, an $\mathbb{F}$ vector space as: $C_n(e_i) = \sum_{j=1}^n e_{i,j}$.

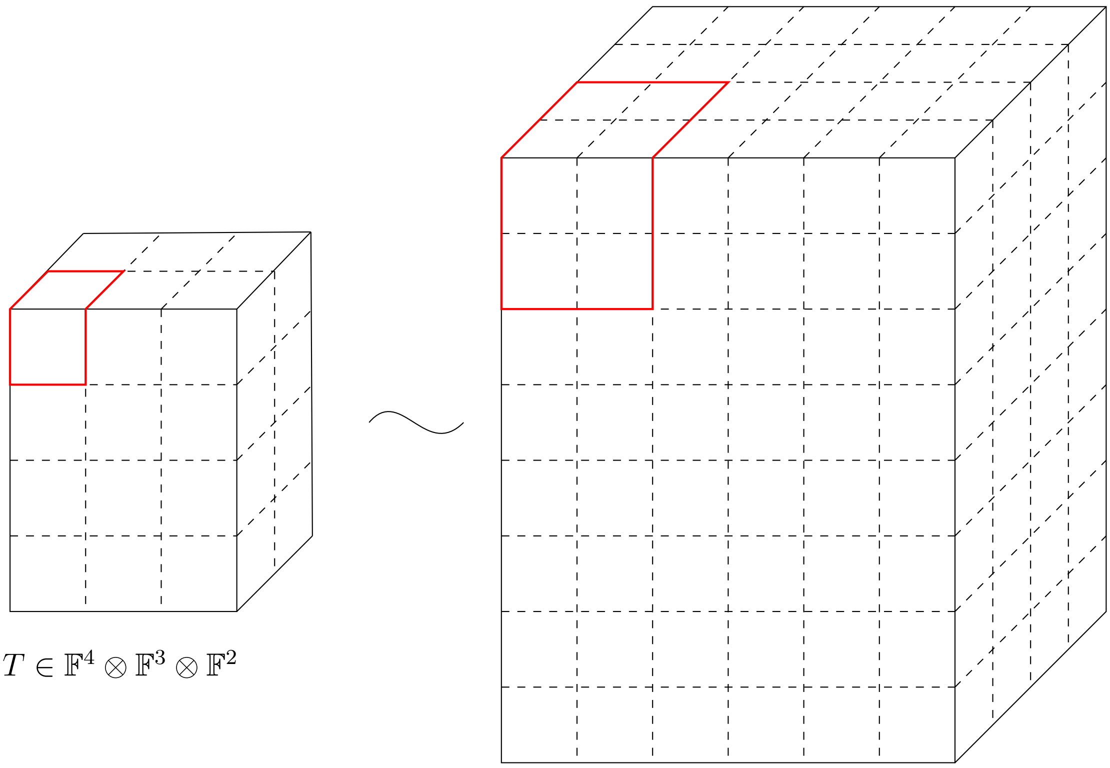

Figure 8: 2 clone of a tensor T

Lemma 4.3. *The cloning operation as defined above distributes over the tensor product i.e. $C_n(A \otimes B) = C_n(A) \otimes C_n(B)$ where A, B are vectors in an arbitrary vector space.*

Proof. Consider two arbitrary vectors, $A := (a_1, \dots, a_k), B := (b_1, \dots, b_k) \in V$, an $\mathbb{F}$ vector space. Then $A \otimes B$ is an order 2 tensor (a matrix) lying in $V \otimes V$, and $(A \otimes B)(i, j) = a_i b_j$ Then according to Definition 4.6,

$$C_n(A \otimes B)(k_1, k_2) = (A \otimes B)(k_1 (\bmod n), k_2 (\bmod n)) = A(k_1 (\bmod n))B(k_2 (\bmod n)),$$

here using the regular modulo operation from number theory. This tensor lies in $(\bigoplus_{i=1}^n V) \otimes (\bigoplus_{i=1}^n V)$.

Now taking the vectors $C_n(A), C_n(B) \in (\bigoplus_{i=1}^n V)$. Here

$$C_n(A) = (\underbrace{a_1, \dots, a_1}_{n \text{ times}}, \dots \underbrace{a_k, \dots, a_k}_{n \text{ times}}),$$

and likewise for $C_n(B)$. Their tensor product is also a matrix lying in $(\bigoplus_{i=1}^n V) \otimes (\bigoplus_{i=1}^n V)$ where now

$$(C_n(A) \otimes C_n(B))(k_1, k_2) = C_n(A)(k_1) C_n(B)(k_2) = A(k_1 (\bmod n))B(k_2 (\bmod n)).$$

Thus, since each component of both tensors is equal, they are the same tensor. □

Remark. For two tensors A, B to show that cloning distributes over the tensor product $A \otimes B$ is a little more involved. However this problem simplifies to asking whether

$C_n(a \otimes b \otimes c) = C_n(a) \otimes C_n(b) \otimes C_n(c)$ where a, b, c are vectors. This simplified problem can be solved in the same way as the proof above, by working in coordinates and using the modulo function from number theory.

Another description of cloning is by first transforming the tensor under consideration in the following manner. Denote $\mathbf{1}_n$ as the vector of 1's lying in $\mathbb{F}^n$. Then the n-clone of a simple vector v can be described as $(v \otimes \mathbf{1}_n)^{\emptyset}$ or the flattening of the matrix created by tensoring v with the vector of ones. Thus another way to look at $C_n(T)$ is with the following procedure:

$$\begin{aligned}
C_n(T) &= C_n\left(\sum_{i=1}^{r} T_i\right) \\
&= \sum_{i=1}^{r} C_n(T_i) \\
&= \sum_{i=1}^{r} C_n(v_1^i \otimes \cdots \otimes v_d^i) \\
&= \sum_{i=1}^{r} C_n(v_i^i) \otimes \cdots \otimes C_n(v_d^i) \\
&= \sum_{i=1}^{r} (v_1^i \otimes \mathbf{1}_n)^{\emptyset} \otimes \cdots \otimes (v_d^i \otimes \mathbf{1}_n)^{\emptyset}
\end{aligned}$$

Remark. This property, along with the observation that cloning is a linear operation, allows for the proof that cloning a tensor does preserve its rank.

Proposition 4.2. ***Rk***$(T) =$ ***Rk***$(C_n(T))$ *for C_n the cloning operation as defined in Definition 4.6, and $T \in V := V_1 \otimes \cdots \otimes V_d$.*

Proof. The proof is straightforward, and is as follows

$\leq$: Since T can be seen as a sub-tensor lying in $C_n(T)$, this inequality naturally exists. The rank of a subtensor is always at most the rank of the tensor, but never more.

$\geq$: Since cloning distributes over addition (Lemma 4.3), if one has the rank decomposition $T = \sum_{i=1}^{r} T_i$, then $C_n(T) = \sum_{i=1}^{r} C_n(T_i)$ is a decomposition of $C_n(T)$. Thus the rank of $C_n(T)$ is at most that of T.

□

4.2 Examining the Counterexample over $\mathbb{R}$

Wang and Seigal [26] in 2023 published a paper that demonstrated the construction of a distinct counterexample to Comon's conjecture, following Shitov [22]. Their paper first examined the constructions made by Shitov, established new notation and terminology to prove results on rank pre- and post-operations, and finally they used MATLAB to construct a distinct order-6 example with real rank 30913 and real symmetric rank 30914.

As the title of their paper suggests, constructing such a counterexample relies on certain lower bounds and inequalities remaining strict, particularly when making constructions in a field, namely $\mathbb{R}$, that is not algebraically complete.

4.2.1 Details from the Paper

The paper in 2023 [26] works over $\mathbb{R}$, specifically in $\mathbb{R}[x, y]$ as the underlying ring of polynomials from which the homogenous polynomials can be selected. For simplicity's sake, there are only 2 variables, corresponding to an underlying real vector space of dimension 2.

The particular tensor under consideration is represented by the homogenous polynomial $x^d - ky^d$ where $d > 0$ is a fixed, even integer, and $k > 0$ (also fixed). Such a homogenous polynomial corresponds to the following tensor

$$T := (e_1)^{\otimes d} - k(e_2)^{\otimes d}, k > 0, d = 2n, n \in \mathbb{Z}_+ \tag{4.1}$$

Visualised as a multiway array, this is a tensor that has non-zero entries at the two extreme "corners" and 0's everywhere else. Such a tensor clearly has rank and symmetric rank as 2.

Since the stated goal is to construct a counterexample to Comon's conjecture, this initial "seed" tensor, T, is first modified by the using a specific family of slices to generate a linear span of tensors (see Section 4.1.3). Such a family when constructed has elements having minimal rank and symmetric rank. The elements in this family achieiving these minimal values do not have to be the same.

The family chosen in this case is

$$\mathcal{M} := \{x^{d-1-i}y^i\}_{i=1}^{d-2} \tag{4.2}$$

This is a collection of monomials of degree $(d-1)$ that are of *mixed* form i.e. they are monomials not purely in either variable. When describing such a family in terms of symmetric tensors (see Section 2.3), the family can be written as

$$\mathcal{M} := \left\{ \frac{1}{(d-1)!} \sum_{\sigma \in S_{d-1}} \sigma((e_1)^{\otimes d-1-i} \otimes (e_2)^{\otimes i}) \right\}_{i=1}^{d-2} \tag{4.3}$$

These order-$(d-1)$ tensors, since they are not formed of a single standard ordered basis vector, they do not affect the initial tensor T, when performing the operation of generating a family modulo slices: $T \bmod \mathcal{M}$. (Note here since they are working with symmetric tensors, it is enough to specify only one family.)

Establishing A Difference in Rank and Symmetric Rank

Within the constructed family $T \bmod \mathcal{M}$, a tensor $P \in T \bmod \mathcal{M}$ corresponds to the following polynomial

$$P(x, y) := x^d - ky^d + \sum_{i=1}^{d-2} (\alpha_i x + \beta_i y)x^{d-1-i}y^i, \alpha_i, \beta_i \in \mathbb{R}\ \forall\ i = 1, \ldots, d-2, k > 0 \tag{4.4}$$

First, the symmetric rank decomposition is computed, and shown to be at least 2 or higher.

Proposition 4.3 ([26]). *A polynomial of the form $P(x, y)$ (as Equation (4.4)) has symmetric rank at least 2.*

Proof. Suppose not. Then one can rewrite $P(x, y)$ in terms of a linear polynomial raised to the dth power i.e.

$$P(x, y) = (ax + by)^d$$

for some choices of $a, b \in \mathbb{R}$.
Expanding the RHS would then lead to the following

$$\begin{aligned} P(x, y) &= (ax + by)^d \\ &= \sum_{j=0}^{d} \binom{d}{j} a^{d-j} b^j x^{d-j} y^j. \end{aligned}$$

This leads to the following equality of polynomial expressions:

$$x^d - ky^d + \sum_{i=1}^{d-2} (\alpha_i x + \beta_i y) x^{d-1-i} y^i = \sum_{j=0}^{d} \binom{d}{j} a^{d-j} b^j x^{d-j} y^j.$$

This leads to a contradiction simply by looking at the coefficient of the y^d monomial on either side of the equality. One must have $b^d = -k$ for some choice of $b \in \mathbb{R}$. Since such an equation has no real solutions, the initial assumption of being able to write P in terms of the d-th power of a single linear form was false, and therefore the Waring rank (equivalent to the symmetric rank) of $P \geq 2$. □

This establishes $\min_{P \in T \bmod \mathcal{M}} \textbf{SRk}(P) = 2$. The next step in the paper is the establishing of the fact that the minimum *rank* element of such a family has rank 1.

Proposition 4.4 ([26]). *There is a tensor of the form $P(x, y)$ (see Equation (4.4)) that is an elementary tensor*

Proof. Consider the original tensor $T := x^d - ky^d$ (see Equation (4.1)). This is a tensor of the form as in Equation (4.4). Now summing all the elements in $\mathcal{M}$ as defined earlier (see Equation (4.2)) gives an order $d - 1$ tensor with non-zero entries everywhere except for two "corners" — corresponding to x^{d-1} and y^{d-1}. These can be added along the d-th mode to the first slice, and after scaling by a factor of $-k$, to the last slice.

Formally the tensor T is now modified to

$$x^d - ky^d + \underbrace{\sum_{i=1}^{d-2} x^{d-i} y^i}_{\text{modifying the first } d\text{-slice}} + \underbrace{-k \left(\sum_{j=1}^{d-2} x^{d-1-j} y^{j+1} \right)}_{\text{scaling and modifying the last } d\text{-slice}}$$

An important point to note here is that changing the degree of the polynomial changes the order, and since the tensors all lie in $(\mathbb{R}^2)^{\otimes d}$, changes affecting the "first" slice have an extra factor of x, and changes affecting the "last" or second have an extra factor of y.

The final step of construction is adding the order-$(d-1)$ tensor corresponding to the monomial $x^{d-2}y$ (scaled by an appropriate a_i) to the first i-slice for all d modes of the tensor T. Likewise for the tensor corresponding to the monomial xy^{d-2} (with appropriate b_i added to the last i-slices). The choices of these scaling coefficients are such that the following conditions are satisfied:

$$\left(\sum_{i=1}^{d} a_i\right) - a_j = 0, \forall j \neq d \quad (4.5)$$

$$\sum_{i=1}^{d-1} a_i = -k \quad (4.6)$$

$$\left(\sum_{i=1}^{d} b_i\right) - b_j = 0, \forall j \neq d \quad (4.7)$$

$$\sum_{i=1}^{d-1} b_i = 1 \quad (4.8)$$

These choices for a_i are made because modifying the tensor by adding the scaled tensors modifies only one entry in last d slice. The last addition of $x^{d-2}y$ along the d-mode only affects the first d-slice. A similar line of reasoning follows the choices for the b_i. Making such choices is always possible, and non-unique.

Performing these operations results in a tensor still lying in the family $T \bmod \mathcal{M}$, now of the form $(\mathbf{1}_2)^{\otimes d-1} \otimes (1, -k)$, an elementary order d tensor. Here again $\mathbf{1}_n$ denotes the vector of all 1's lying in $\mathbb{F}^n$ □

Remark. It is useful to note that proving such an elementary tensor exists is not a function of the underlying field, assuming characteristic 0. Such a proof could also be modified for any number of variables (moving from working in $\mathbb{F}^2$ to $\mathbb{F}^n$), and for tensors of any even order $d > 0$.

The following statement holds for the family $T \bmod \mathcal{M}$

$$\min \mathbf{Rk}(T \bmod \mathcal{M}) < \min \mathbf{SRk}(T \bmod \mathcal{M}) \quad (4.9)$$

Moving from Families to a Specific Tensor

Now that the existence of a family having elements with differing minimal rank and symmetric rank has been established, the authors construct $SAdj(T, \mathcal{M})$, as a symmetric tensor that is a candidate counterexample to Comon's conjecture. The problem of computing the rank of such a construction now arises, as it relates to the original T and candidate family $\mathcal{M}$.

The solution proposed originally by Shitov [22], is a modification of $\mathcal{M}$ to a family of *elementary* order-$(d-1)$ tensors that span the same linear space. Unfortunately, in the format $(\mathbb{R}^2)^{\otimes 4}$, doing so also violates the vital inequality required for rank and symmetric rank of the family $T \bmod \mathcal{M}$ (see Equation (4.9)).

Such a modification is required for the original computation of rank to be an equality (see Theorem 4.1), because equality only holds when the family consists of elementary tensors.

The following proposition is adapted from Proposition 4.2 in [26]. It shows that for orders $d = 2n, n > 2$, modifying the initial family $\mathcal{M}$ to a finite family of elementary tensors $\mathcal{W}$ is possible while maintaining the inequality eq. (4.9). This argument holds specifically for the order 6, though it does not appear to be addressed in their paper.

Proposition 4.5. *Consider T as in Equation (4.1), and $\mathcal{M}$ as Equation (4.2). Let $\mathcal{W} \subset (\mathbb{R}^2)^{\otimes d-1}$, a finite set of elementary symmetric tensors such that $\mathrm{span}\ \mathcal{W} \supseteq \mathcal{M}$. Then $0 \notin T \,\mathrm{mod} \mathcal{W}$ when $d = 2n, n > 2$.*

Proof. Consider $\mathcal{W}$, a collection of elementary symmetric tensors of appropriate order. Suppose $0 \in T \,\mathrm{mod}\, \mathcal{W}$. This is equivalent to asking that $\mathcal{W}$ spans the set $\{x^i y^{d-i}\}_{i=0}^{d-1} \supseteq \mathcal{M}$. In particular, if one can show that the dimension of span $\mathcal{W}$ is d (the number of degree $d-1$ monomials in 2 variables), this completes the proof.

Consider the tensor represented by $x^{d-2}y$. This monomial has rank $d-1$ since it can be written minimally as sum of $d-1$ elementary tensors. Thus the dimension of span $\mathcal{W}$ is at least $d-1$. Since $\mathcal{W}$ consists of symmetric tensors, its basis vectors must be a linear forms raised to the $d-1$th power: $(\lambda_i x + \mu_i y)^{d-1}$. These forms must also correspond to elementary symmetric tensors, by virtue of being elements of $\mathcal{W}$.

Now since this set, $\{(\lambda_i x + \mu_i y)^{d-1}\}_{i=1}^{d-1}$ is a basis for span $\mathcal{W} \supseteq \mathcal{M}$, one must have that the tensors $x^{d-2}y$ and xy^{d-2} can be written as linear combinations of the elements of the set. Doing so imposes conditions on $\lambda_i, \mu_i > 0$ such that

$$\sum_{i=1}^{d} \frac{\lambda_i}{\mu_i} = 0 = \sum_{i=1}^{d} \frac{\mu_i}{\lambda_i}.$$

This set of equalities results in the statement

$$1 = \frac{\lambda_1}{\mu_1}\frac{\mu_1}{\lambda_1} = \left(\sum_{i=2}^{d} \frac{\lambda_i}{\mu_i}\right)\left(\sum_{i=2}^{d} \frac{\mu_i}{\lambda_i}\right) = (d-2) + \sum_{i=2}^{d}\sum_{j\neq i} \frac{\lambda_i \mu_j}{\mu_i \lambda_j}. \tag{4.10}$$

If one can show that

$$\sum_{i=2}^{d}\sum_{j<i} t_{ij} + t_{ij}^{-1} + d - 3 \neq 0, \tag{4.11}$$

where $t_{ij} := \frac{\lambda_i \mu_j}{\mu_i \lambda_j}$, then equation 4.10 is false. However, the range of the function, seen in equation 4.11, is the entirety of $\mathbb{R}$ whenever dealing with a value of $d > 4$ since the function generated in terms of $\{t_{ij}, t_{ij}^{-1}\}$ is no longer univariate.

Therefore there are choices of λ_i, μ_i that can be made such that the span of $\mathcal{W}$ has a dimension less than d. This is a contradiction to the idea that $\mathcal{W}$ spans the set $\{x^i y^{d-i}\}_{i=0}^{d-1}$, containing d linearly independent elements. Thus the initial assumption of 0 lying in $T \,\mathrm{mod}\, (\mathcal{W})$ was false.

□

4.2.2 Emulating Family Construction

While for the order-4 case from Shitov, the issue of Equation (4.9) being violated when modifying the family $\mathcal{M}$ arose, as seen in the previous proposition, such issues do not arise when working with the order-6 or higher cases. However, the authors of [26] continue to use the cloning methodology later in their paper, with little explanation as to why it is still necessary to do so. Cloning does provide more variables, allowing for situations where modifying $\mathcal{M}$ to $\mathcal{W}$ causes changes to the inequality $\min \textbf{Rk}(T \bmod (\mathcal{M})) < \min \textbf{SRk}(T \bmod (\mathcal{M}))$ back into $\min \textbf{Rk}(T \bmod (\mathcal{W})) \leq \min \textbf{SRk}(T \bmod (\mathcal{W}))$.

As demonstrated in Section 4.1.4, since cloning preserves rank, then inequality is preserved after taking the k-th clones, and transforms into

$$\min \textbf{Rk} C_k(T) \bmod C_k(\mathcal{M}) < \min \textbf{SRk} C_k(T) \bmod C_k(\mathcal{M}) \tag{4.12}$$

In this situation, modifying $C_k(M)$ (a family of order-$(d-1)$ tensors lying in $C_k(V^{\otimes d-1})$) into $\mathcal{W}$ (a family of order-$(d-1)$ elementary tensors) can occur successfully. This results in working with a symmetric tensor $SAdj(C_k(T), \mathcal{W})$ - the final symmetric tensor having a definite distinction between rank and symmetric rank. In all the papers surveyed that cover this cloning methodology [22, 24, 26], the choice of k is made such that conditions detailed in Propositions 4.6 and 4.7 from [26] are valid.

It is past this point that both Shitov [22], and Wang and Seigal [26] go forward with the construction of a family $\mathcal{W}$ to replace $C_k(\mathcal{M})$ that can be described algorithmically, but the justification for why such choices are made remains opaque. Personal correspondence with Shitov confirms the idea that such families can be constructed, but a general procedure for any even dimension is still elusive.

The later paper [26] constructs this family for their candidate counterexample of order 6. The elements of this familiy are described below for the sake of completeness. In personal correspondence, the authors indicated that this falls in step with the construction in the preprint [22]:

They begin with vectors lying in $\mathbb{R}^{14}$ of the form a_i, where $i \in [n = 7]$ and each vector is such that $a_i(2i) = a_i(2i-1) = 1$ and all the other entries are 0. Then the new vectors constructed lie in $\mathbb{R}^{28}$ and are one of the following forms (where the | indicates the distinction between the first 14 entries and the last 14 entries.)

1. $(a_{i_1} + a_{i_2} + a_{i_3} + a_{i_4} \mid a_{k_1})$
2. $(a_{i_1} + a_{i_2} + a_{i_3} + a_{i_4} \mid 0)$
3. $(a_{i_1} + a_{i_2} + a_{i_3} \mid \frac{n-2}{n-1} a_{k_1})$
4. $(a_{i_1} + a_{i_2} + a_{i_3} \mid \frac{n-3}{n-4} a_{k_1})$
5. $(a_{i_1} + a_{i_2} + a_{i_3} \mid a_{k_1} + a_{k_2})$
6. $(a_{i_1} + a_{i_2} + a_{i_3} \mid 0)$
7. $(a_{i_1} + a_{i_2} \mid \frac{(n-2)^2}{(n-3)(n-1)} a_{k_1})$
8. $(a_{i_1} + a_{i_2} \mid \frac{n-2}{n-4} a_{k_1})$

9. $(a_{i_1} + a_{i_2} \mid \frac{n-2}{n-3}(a_{k_1} + a_{k_2}))$
10. $(a_{i_1} + a_{i_2} \mid 0)$
11. $(a_{i_1} \mid \frac{n-2}{n-3} a_{k_1})$
12. $(a_{i_1} \mid \frac{n-1}{n-4} a_{k_1})$
13. $(a_{i_1} \mid \frac{n-1}{n-3}(a_{k_1} + a_{k_2}))$
14. $(a_{i_1} \mid 0)$
15. $(0 \mid a_{k_1})$
16. $(0 \mid a_{k_1} + a_{k_2})$

Here $1 \leq i_1 < i_2 < i_3 < i_4 \leq 7$ and $1 \leq k_1 < k_2 \leq 7$. The collection of all the 16 classes of vectors in $\mathbb{R}^{28}$ described above, for all choices of indices i_1, i_2, i_3, i_4 and k_1, k_2 forms the set $\mathcal{W}_1$. This is only half of their construction.

Under the permutation operation

$$\pi : (i_1, i_2, \ldots, i_{14}, k_1, k_2, \ldots, k_{14}) \mapsto (k_2, \ldots, k_{14}, k_1, i_2, \ldots, i_{14}, i_1),$$

the same set $\mathcal{W}_1$ can be transformed into a new set $\mathcal{W}_2$. The final emulating family they use for their order-6 counterexample is $\mathcal{W} := \mathcal{W}_1 \cup \mathcal{W}_2$

This emulating family proves to be the last piece in their construction of $SAdj(C_k(T), \mathcal{W})$, an order-$6$ symmetric tensor having real symmetric rank 30914 and real rank 30913.

Chapter 5

Conclusion

5.1 Results of this Thesis

While this thesis only condenses the literature surrounding Comon's conjecture and attempts to generate counterexamples to it, the fact that a computed, real counterexample does exist provides some hope for a conclusive answer in the future. As mentioned in Chapter 4, since the real counterexample requires the discrepancy from Equation (4.9), an equivalent condition in the complex case seems to be the pathway forward.

To be more precise: if a symmetric, complex-valued, order-d tensor T exists, along with a family of symmetric, complex-valued, order-$(d-1)$ tensors $\mathcal{M}$, such that the statement

$$\min \textbf{Rk}(T \operatorname{mod} \mathcal{M}) < \min \textbf{SRk}(T \operatorname{mod} \mathcal{M})$$

still holds, then a viable candidate counterexample can be constructed, paralleling the constructions in [22, 26].

This thesis also provides mathematical definitions for the operations of adjoining, modulo and cloning that are referenced in the literature, but have their definitions only in terms of the d-way arrays.

Finally, even though [26] is now the only published counterexample to Comon's conjecture, Proposition 4.5 in this thesis shines light on the fact that the cloning operation may not be as necessary for symmetric tensors of order $d = 2n, n > 2$. While correspondence has confirmed that counterexamples at least in the order-4 and 6 case can be computed, the idea that there is no general generating procedure is less satisfactory to accept.

5.2 Future Work

The work in this thesis establishes a baseline for exploring other constructions of candidate counterexamples to Comon's Conjecture.

- One way this work can be extended is to focus more on a geometric description of the constructions made in this paper, particularly for Equation (4.9). In doing so, the hope is that one would uncover geometric conditions that indicate a candidate counterexample to the conjecture, and these are hopefully independent of the base field chosen. [ILLUSTRATE/ELABORATE?]

- A more explicit understanding of the choices of vectors made in Section 4.2.2. The choices made so far stem from [22], and while they have proved to be effective, have no indication as to *why* they are. Studying the construction of such a family more carefully might indicate algebraic or geometric conditions that are being met by the collection of vectors or slices, allowing for the existence of an emulating family of elementary tensors.

- Comon's conjecture as originally stated in [18] was only for $\mathbb{R}$- or $\mathbb{C}$-valued tensors. A conclusive understanding of whether Comon's conjecture can be extended to finite fields, or if such an extension is non-trivial can be another path forward.

- Adjusting the work in this paper to understand the smallest possible cloning operation required to satisfy the conditions of the emulating family could be another potential extension of this work. The paper from Borovik et al. in 2025 [5], while not explicitly related to Comon's conjecture, does establish an upper bound for how large a cloning operation may be required. This paper builds on [23], a preprint from Shitov more in line with constructing a counterexample to Strassen's conjecture, but relies on similar methodology.

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